Reading Notes: Analytic Combinatorics

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Contents

Chapter 1

COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.*

A unimodal permutation climbs to a single peak and then descends. In symbolic terms,

$$\mathcal{P}^{\mathcal{U}} = \textsc{Set}(\mathcal{Z}) \star \mathcal{Z}^{\blacksquare} \star \textsc{Set}(\mathcal{Z})$$
$$P^U(z) = \int_0^z \left(e^t \cdot \frac{\mathrm{d}}{\mathrm{d}t} t \cdot e^t\right) \mathrm{d}t = \frac{1}{2}(e^{2z} - 1)$$

which immediately yields

$$P_n^U = n![z^n]\frac{1}{2}e^{2z} = n!\frac{1}{2}\frac{2^n}{n!} = 2^{n-1}$$

For a direct counting argument,

$$P_n^U = \sum_{k=1}^{n} \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}$$

where k is the position of n in the permutation.

1.2 Admissible constructions and specifications

1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.*

Consider partitions of size n with at most k summands, each bounded by l. Their generating function is

$$[z^n]\frac{(1-z)(1-z^2)\cdots(1-z^{k+l})}{((1-z)(1-z^2)\cdots(1-z^k)) \cdot ((1-z)(1-z^2)\cdots(1-z^l))} \tag{1.1}$$

Let $f(k,l)$ denote this generating function, so the coefficient of z^n counts the admissible partitions of n. The natural split is by the largest part: it is either strictly smaller than l, or exactly l. Thus

$$f(k,l) = f(k,l-1) + z^l f(k-1,l) \tag{1.2}$$

To package the dependence on l, write

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l$$

For the base case $k=0$, only the empty partition survives, so $f(0,l)=1$ for all $l \geq 0$. Hence

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Now multiply 1.2 by x^l and sum over $l \geq 1$:

$$\sum_{l=1}^{\infty} f(k,l)x^l - \sum_{l=1}^{\infty} f(k,l-1)x^l = \sum_{l=1}^{\infty} f(k-1,l)(zx)^l$$

- The first term simplifies to $F_k(x) - 1$, since $f(k,0)=1$.
- The second term is simply $xF_k(x)$.
- The right-hand side yields $F_{k-1}(zx) - 1$.

Substituting these pieces back in gives

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x}F_{k-1}(zx)$$

Iterating the relation gives

$$\begin{aligned} F_k(x) &= \frac{1}{1-x}F_{k-1}(zx) \\ &= \frac{1}{1-x}\frac{1}{1-zx}F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx)\cdots(1-z^{k-1}x)}F_0(z^kx) \end{aligned}$$

Since $F_0(x) = \frac{1}{1-x}$, we have $F_0(z^kx) = \frac{1}{1-z^kx}$. Thus

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l = \frac{1}{(1-x)(1-zx)\cdots(1-z^kx)}$$

Since the denominator has distinct linear factors, we may use partial fractions:

$$\frac{1}{\prod_{i=0}^{k}(1-z^ix)} = \sum_{j=0}^{k} \frac{A_j}{1-z^jx}$$

To isolate A_j, multiply through by $(1 - z^j x)$ and then set $x = z^{-j}$:

$$\begin{aligned} A_j &= \left[\frac{1 - z^j x}{\prod_{i=0}^{k} (1 - z^i x)} \right]_{x = z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1} (1 - z^i z^{-j}) \prod_{i=j+1}^{k} (1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \end{aligned}$$

The coefficient of x^l in $F_k(x)$ is then read off term by term. For $\frac{A_j}{1 - z^j x}$, it is simply $A_j (z^j)^l$.

$$\begin{aligned} f(k,l) &= \sum_{j=0}^{k} A_j (z^j)^l \\ &= \sum_{j=0}^{k} \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^{k} (1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^{k} (1 - z^i)}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

A small rearrangement of the exponent gives

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

so the sum becomes

$$f(k,l) = \frac{1}{\prod_{i=1}^{k} (1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The q-binomial theorem states that for a variable X:

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \tag{1.3}$$

Substituting $X = z^{l+1}$ into the theorem recovers 1.1:

$$\begin{aligned} f(k,l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

For completeness, let us also justify 1.3. Set

$$P_k(X) = \prod_{i=0}^{k-1}(1 - z^i X)$$

We claim that $P_k(X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$. Start from the elementary relation

$$P_k(X) = (1 - z^{k-1}X)P_{k-1}(X)$$

Next, replace X by zX:

$$P_k(zX) = \prod_{i=0}^{k-1}(1 - z^{i+1}X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

After rearranging, this becomes the functional equation

$$(1 - X)P_k(zX) = (1 - z^k X)P_k(X)$$

Let $P_k(X) = \sum_{j=0}^{k} c_j X^j$. We substitute this power series into the functional equation:

$$(1 - X)\sum_{j=0}^{k} c_j (zX)^j = (1 - z^k X)\sum_{j=0}^{k} c_j X^j$$

Comparing coefficients of X^j on both sides yields:

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

so

$$c_j = c_{j-1}\frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1} \frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from $c_0 = P_k(0) = 1$, we iterate:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1} \frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of z is $0 + 1 + 2 + \cdots + (j - 1) = \frac{j(j-1)}{2}$, and the remaining fraction is exactly the Gaussian binomial coefficient. Therefore,

$$c_j = (-1)^j z^{j(j-1)/2} \binom{k}{j}_z$$

Plugging the coefficients back into the expansion, we obtain

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

which is exactly what we wanted.

$\triangleright$ **I.17.** *Systems of linear Diophantine inequalities.*

This note contains five separate counting problems. They are related by theme, but each one has its own constraint set and should be read independently.

Problem 1

$$\{x_1 \geq 0,\ x_2 \geq 2x_1,\ x_3 \geq 2x_2,\ x_4 \geq 2x_3\}$$

At first glance this system looks mildly unfriendly, but slack variables settle it almost at once. Introduce non-negative variables $y_j \in \mathbb{Z}_{\geq 0}$ to measure the "excess" of each term above its lower bound.

$$\begin{aligned}
x_1 &= y_1 \\
x_2 &= 2x_1 + y_2 = 2y_1 + y_2 \\
x_3 &= 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \\
x_4 &= 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4
\end{aligned}$$

The total weight is:

$$S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4.$$

Now the generating function is immediate: the variables y_j are independent and each ranges freely over the non-negative integers, so the OGF $F(z)$ is just the product of the four one-variable OGFs:

- For $15y_1 : \sum_{y_1=0}^{\infty}(z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For $7y_2 : \sum_{y_2=0}^{\infty}(z^{7})^{y_2} = \frac{1}{1-z^{7}}$
- For $3y_3 : \sum_{y_3=0}^{\infty}(z^{3})^{y_3} = \frac{1}{1-z^{3}}$
- For $y_4 : \sum_{y_4=0}^{\infty}(z^{1})^{y_4} = \frac{1}{1-z^{1}}$

Multiplying them together gives the answer:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

Problem 2

$$\{x_1 + x_2 \leq x_3\}$$

This is a new problem, so we start again from its own constraint. Here slack variables turn the inequality into an equality with independent parts. The constraint $x_3 \geq x_1 + x_2$ can be rewritten by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y$$

The size of the composition is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables x_1, x_2, y, x_4 are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For $2x_1 : \frac{1}{1-z^2}$
- For $2x_2 : \frac{1}{1-z^2}$
- For $y : \frac{1}{1-z}$
- For $x_4 : \frac{1}{1-z}$

Multiplying these factors gives

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Problem 3

$$\{x_1 + x_2 \geq x_3\}$$

This third problem is independent of the previous two. Since the inequality does not suggest a single slack variable, the cleanest route is to work with the complement. Let $U(z) = (1-z)^{-4}$ be the OGF for four unrestricted variables. The universe of compositions partitions into those where $x_1 + x_2 < x_3$ and our target class $x_1 + x_2 \geq x_3$. We find the OGF for the complement $x_1 + x_2 < x_3$. Set $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum S is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, denoted by $C(z)$, is

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

Problem 4

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

We now move to a fourth, separate problem. To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, combine the complement method with the symmetry of the variables. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all compositions (x_1, x_2, x_3, x_4), there are three mutually exclusive cases 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since A and B are built from the same number of variables with the same weights. Denote this common OGF by $G_<(z)$, and let $G_=(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \frac{1}{(1-z)^4}$. Therefore,

$$U(z) = 2G_<(z) + G_=(z)$$

Our target OGF is $F(z) = G_<(z) + G_=(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_=(z)}{2}$$

Now $G_=(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is: $[z^n]\frac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty}(n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2}\left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3}\right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

Problem 5

$$\{x_1 \leq x_2,\ x_2 \geq x_3,\ x_3 \leq x_4\}$$

The fifth problem is again independent. Because the inequalities alternate direction, we cannot use the simple slack-variable chain that works for $x_1 \leq x_2 \leq x_3 \leq x_4$. Instead, we use inclusion–exclusion. We begin with the universe of all unrestricted compositions, whose OGF is $U(z) = \frac{1}{(1-z)^4}$. We wish to enforce three conditions

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

It is easier to work with chains of "less than or equal" inequalities, so consider the complement of the middle downward step. Take the set of all compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. From this set, we subtract those compositions that violate the middle constraint (i.e., $x_2 < x_3$).

The OGF for the set defined by $x_1 \leq x_2$ and $x_3 \leq x_4$ is easy to obtain because the two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at x_2 fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)}\left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)}\right] \end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.*

This bijection looks clever on first meeting and then almost inevitable in retrospect. We build a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd → Distinct
  1. Group identical odd parts together.
  2. If an odd part k appears m times, write m in binary $m = 2^{a_1} + 2^{a_2} + \cdots$ with distinct powers of two.
  3. Replace the m copies of k by $2^{a_1} \cdot k,\ 2^{a_2} \cdot k,\ \ldots$
- Distinct → Odd
  1. Factor each part as $2^a \times$ odd.
  2. Split it into 2^a copies of that odd factor.
  3. Do this for all parts; now all parts are odd.
  4. Sort the resulting parts to obtain a partition into odd parts.

Here is a concrete pass through the map:

$$\begin{aligned} 3+3+3+3+3+1+1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned}$$

$$\begin{aligned}12 + 3 + 2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3 + 3 + 3 + 3) + 3 + (1 + 1) \\ &= 3 + 3 + 3 + 3 + 3 + 1 + 1\end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.*

To get a usable recurrence for the partition numbers, it is natural to look at the reciprocal of $P(z)$:

$$A(z) = \prod_{n=1}^{\infty}(1 - z^n)$$

Euler's Pentagonal Number Theorem says that this product collapses to a remarkably sparse series:

$$\begin{aligned}A(z) &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots\end{aligned}$$

The exponents $g_k = \frac{k(3k-1)}{2}$ are the generalized pentagonal numbers. Since $A(z)$ is the reciprocal of $P(z)$, we have

$$\sum_{n=0}^{\infty} p(n) z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing coefficients of z^n now gives the recurrence that is genuinely useful in computation:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots$$

A straightforward Python implementation is:

```
def compute_partitions(N):
P = [0] * (N + 1)
P[0] = 1  # Base case: p(0) = 1

for i in range(1, N + 1):
    k = 1
    while True:
        # Calculate pentagonal numbers for k and -k
        g_k_pos = k * (3 * k - 1) // 2
        g_k_neg = k * (3 * k + 1) // 2

        # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k=3,4 (terms 3,4)...
        # Simplest way: (-1)^(k-1)
        sign = 1 if k % 2 == 1 else -1

        # Add p(i - g_k_pos)
        if g_k_pos <= i:
            P[i] += sign * P[i - g_k_pos]
        else:
            break # No more terms will fit

        # Add p(i - g_k_neg)
        if g_k_neg <= i:
            P[i] += sign * P[i - g_k_neg]
        else:
            break

        k += 1
return P
```

▷ **I.20.** *A digital surprise.*

$$
\begin{aligned}
\varphi &= (1-10^{-1})(1-10^{-2})(1-10^{-3})(1-10^{-4})\cdots \\
&= \prod_{n=1}^{\infty}(1-z^n)\Big|_{z=.1} \\
&= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + z^{22} + z^{26} - \cdots\big|_{z=.1} \\
&= .\dot{9} - .1 - .01 + .00001 + .0000001 - .000000000001 - \cdots
\end{aligned}
$$

The partial sums are

- $\varphi_1 = .9999999999999999999999999$
- $\varphi_2 = .8999999999999999999999999$
- $\varphi_3 = .8899999999999999999999999$
- $\varphi_4 = .8900099999999999999999999$

- $\varphi_5 = .890010099999999999999999$
- $\varphi_6 = .890010099998999999999999$
- $\varphi_7 = .890010099998998999999999$
- $\varphi_8 = .890010099998999000000999$

▷ **I.21.** *Lattice points.*

We want the number of integer solutions to $\sum_{i=1}^{d} x_i^2 \leq n^2$. The natural object here is the theta series. For one coordinate $x_i \in \mathbb{Z}$, the contribution to the sum of squares is x_i^2, so the one-variable generating function is

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2\sum_{n=1}^{\infty} z^{n^2}$$

Because the coordinates are independent, the generating function for $\sum x_i^2$ is simply the dth power:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

Here $[z^k]S(z)$, denoted $r_d(k)$, counts representations of k as a sum of d squares. Geometrically, that is the number of lattice points on the sphere of radius $\sqrt{k}$. Therefore, the number of points $N(n, d)$ in the closed ball of radius n is:

$$N(n, d) = \sum_{k=0}^{n^2} r_d(k)$$

Passing from exact radius to radius at most n is just a prefix sum, so we multiply by $\frac{1}{1-z}$:

$$\sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

Extracting the coefficient of z^{n^2} finishes the job:

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

1.4 Words and regular languages

▷ **I.23.** *Alice, Bob, and coding bounds.*

By Proposition I.4, the class $\mathcal{C}$ of binary strings avoiding the pattern 11 has generating function

$$C(z) = \frac{1+z}{1-z-z^2}$$

If codewords are allowed to have any length up to L, we need the sum of the first $L+1$ coefficients. Its generating function is:

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}$$

As usual, the nearest singularity controls the asymptotics. Since the pole at $z = 1/\varphi$ (with $\varphi = \frac{1+\sqrt{5}}{2}$) lies closer to the origin than the pole at $z = 1$, we obtain, as $L \to \infty$,

$$\sum_{j=0}^{L} C_j \sim A \cdot \varphi^L$$

for some constant A. To encode n bits of information, we want $2^n \leq \sum_{j=0}^{L} C_j$, so taking base-2 logarithms gives:

$$n \leq \log_2 \left(\sum_{j=0}^{L} C_j \right) = L \log_2 \varphi + O(1)$$

Rearranging for L, we obtain the lower bound:

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

$\triangleright$ **I.27.** *Congruence languages.*

The DFA $\mathcal{A}$ for binary strings congruent to 2 (mod 7) is the standard remainder automaton:

- **States**: A set $\mathcal{Q} = \{S_0, \ldots, S_6\}$, where each state S_r represents the current remainder r modulo 7.
- **Initial State**: S_0 (representing the value 0).
- **Accepting State**: $\overline{Q} = \{S_2\}$.
- **Alphabet**: $\mathcal{A} = \{0, 1\}$.

Each transition follows the familiar update rule $q_r \circ b = q_{(2r+b) \pmod 7}$.

Table 1.1: State Transition

Current State (Remainder)	Input 0 ($2r \pmod 7$)	Input 1 ($2r + 1 \pmod 7$)
S_0	S_0	S_1
S_1	S_2	S_3
S_2	S_4	S_5
S_3	S_6	S_0
S_4	S_1	S_2
S_5	S_3	S_4
S_6	S_5	S_6

$\triangleright$ **I.29.** *The forward equations.*

A last-letter decomposition gives the usual recurrence for the set $\mathcal{M}_k$ of words leading from the initial state q_0 to state q_k:

$$\mathcal{M}_k \cong [\![k = 0]\!] \cdot \{\epsilon\} + \sum_{j,\alpha : q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$

where $[\![\cdot]\!]$ is the Iverson bracket. Translating directly to generating functions,

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}$$

In vector form, with $\mathbf{M}(z)$ as a column vector and $\mathbf{e}_0$ the first basis vector, this is simply the transposed analogue of the backward equations:

$$\mathbf{M}(z)^T = \mathbf{e}_0^T + z\mathbf{M}(z)^T T$$

▷ **I.33.** *Waiting times in strings.*

Let us test the formulas on a deliberately tiny prefix language, $\hat{\mathcal{L}} = \{aa, bbb\}$.

1. Prefix property: aa is not a prefix of bbb, and conversely, so the language is prefix-free.

2. Generating function: the OGF is $\hat{L}(z) = z^2 + z^3$.

3. Probability: $\hat{L}(1/2)$ is the probability that a random infinite string starts with a word of $\hat{\mathcal{L}}$: $\hat{L}(1/2) = (1/2)^2 + (1/2)^3 = 1/4 + 1/8 = 3/8$.

4. Expected time: $\frac{1}{2}\hat{L}'(1/2)$ gives the expected matched length: $\hat{L}'(z) = 2z + 3z^2 \implies \hat{L}'(1/2) = 1.75 \implies E = 0.875$.

5. Unpacking the value:

   - Word "aa": Probability 1/4 of length 2 $\to$ Contribution: $2 \times 0.25 = 0.5$.
   - Word "bbb": Probability 1/8 of length 3 $\to$ Contribution: $3 \times 0.125 = 0.375$.
   - No Match: Probability 5/8 of length 0 $\to$ Contribution: $0 \times 0.625 = 0$.

▷ **I.34.** *A probabilistic paradox on strings.*

Fix a binary pattern of length k. The generating function $T(z)$ for its first occurrence is:

$$T(z) = \frac{z^k}{z^k + (1-mz)c(z)} = \frac{z^k}{z^k + (1-2z)c(z)}$$

The expected waiting time then follows immediately from the derivative at $z = 1/2$:

$$z\frac{\mathrm{d}}{\mathrm{d}z}T(z)\bigg|_{z=1/2} = 2^k c(1/2)$$

1.5 Tree structures

▷ **I.39.** *Unary–binary trees and Motzkin numbers.*

Here the characteristic polynomial is $\phi(u) = 1 + u + u^2$. To extract the number of trees with n nodes, $M_n = [z^n]M(z)$, we apply Lagrange inversion:

$$\begin{aligned}
M_n &= \frac{1}{n}[u^{n-1}]\phi(u)^n \\
&= \frac{1}{n}[u^{n-1}](1+u+u^2)^n \\
&= \frac{1}{n}[u^{n-1}]\sum_j \binom{n}{j}(u+u^2)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-1}]u^j(1+u)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-j-1}]\sum_l \binom{j}{l}u^l \\
&= \frac{1}{n}\sum_j \binom{n}{j}\binom{j}{n-j-1} \\
&= \frac{1}{n}\sum_k \binom{n}{n-k}\binom{n-k}{n-(n-k)-1} \\
&= \frac{1}{n}\sum_k \binom{n}{k}\binom{n-k}{k-1}
\end{aligned}$$

▷ **I.41.** *The Lagrangean form of Schröder's GF.*

What remains is to prove the last equality:

$$\frac{(-1)^{n-1}}{n}\sum_k (-2)^k \binom{n}{k+1}\binom{n+k-1}{k} = \frac{1}{n}\sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k}$$

Multiplying both sides by n simplifies the notation:

$$(-1)^{n-1}\sum_{k=0}^{n-1}(-2)^k \binom{n}{k+1}\binom{n+k-1}{k} = \sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k}$$

Step 1. Re-index the left sum.

Using $\binom{n}{k+1} = \binom{n}{n-1-k}$ and $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$, set $j = n-1-k$. Then

$$L = (-1)^{n-1}\sum_{j=0}^{n-1}(-2)^{n-1-j}\binom{n}{j}\binom{2n-2-j}{n-1} = 2^{n-1}\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j \binom{n}{j}\binom{2n-2-j}{n-1}$$

Step 2. Express as a coefficient extraction.

Since $\binom{2n-2-j}{n-1} = [z^{n-1}](1+z)^{2n-2-j}$, coefficient extraction gives:

$$\begin{aligned}
L &= 2^{n-1}\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j\binom{n}{j}[z^{n-1}](1+z)^{2n-2-j}\\
&= 2^{n-1}[z^{n-1}]\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j\binom{n}{j}(1+z)^{2n-2-j}\\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\sum_{j=0}^{n}\binom{n}{j}\left(-\frac{1}{2(1+z)}\right)^j\\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\left(1-\frac{1}{2(1+z)}\right)^n\\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\frac{(1+2z)^n}{2^n(1+z)^n}\\
&= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(1+2z)^n
\end{aligned}$$

Step 3. Reverse the polynomial.

Let $P(z) = (1+z)^{n-2}(1+2z)^n$, which has degree $2n-2$. The coefficient of z^{n-1} in $P(z)$ is the same as in the reversed polynomial $z^{2n-2}P(1/z)$:

$$z^{2n-2}P(1/z) = z^{2n-2}\left(1+\frac{1}{z}\right)^{n-2}\left(1+\frac{2}{z}\right)^n = (z+1)^{n-2}(z+2)^n$$

Hence,

$$L = \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^n$$

Step 4. Express the right sum as a coefficient.

Likewise, $\binom{2n-k-2}{n-1} = [z^{n-1}](1+z)^{2n-k-2}$, so

$$\begin{aligned}
R &= \sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k}\\
&= \sum_{k=0}^{n-2}[z^{n-1}](1+z)^{2n-k-2}\binom{n-2}{k}\\
&= [z^{n-1}]\sum_{k=0}^{n-2}(1+z)^{2n-k-2}\binom{n-2}{k}\\
&= [z^{n-1}](1+z)^{2n-2}\sum_{k=0}^{n-2}\binom{n-2}{k}(1+z)^{-k}\\
&= [z^{n-1}](1+z)^{2n-2}\left(1+\frac{1}{1+z}\right)^{n-2}\\
&= [z^{n-1}](1+z)^{2n-2}\frac{(2+z)^{n-2}}{(1+z)^{n-2}}\\
&= [z^{n-1}](1+z)^n(2+z)^{n-2}
\end{aligned}$$

Step 5. Show equality of the two coefficient expressions.

So we are down to showing that $\frac{1}{2}(1+z)^{n-2}(2+z)^n$ and $(1+z)^n(2+z)^{n-2}$ have the same coefficient of z^{n-1}. Their difference is:

$$\begin{aligned}\Delta &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^n - [z^{n-1}](1+z)^n(2+z)^{n-2} \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}[(2+z)^2 - 2(1+z)^2] \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}(2-z^2)\end{aligned}$$

Thus it suffices to prove $[z^{n-1}]f(z) = 0$, where

$$f(z) = (1+z)^{n-2}(2+z)^{n-2}(2-z^2)$$

Step 6. A symmetry argument.

$f(z)$ has degree $2n-2$. Now compute $f(2/z)$:

$$\begin{aligned}f(2/z) &= (1+2/z)^{n-2}(2+2/z)^{n-2}\left(2-(2/z)^2\right) \\ &= \left(\frac{2+z}{z}\right)^{n-2}\left(\frac{2(1+z)}{z}\right)^{n-2}\left(2-\frac{4}{z^2}\right) \\ &= \frac{(2+z)^{n-2}(1+z)^{n-2}2^{n-2}}{z^{2n-4}} \cdot \frac{-2(2-z^2)}{z^2} \\ &= -2^{n-1}\frac{f(z)}{z^{2n-2}}\end{aligned}$$

Hence, $z^{2n-2}f(2/z) = -2^{n-1}f(z)$. Writing $f(z) = \sum_{k=0}^{2n-2} a_k z^k$ and comparing the middle coefficient gives: $a_{n-1}2^{n-1} = -2^{n-1}a_{n-1}$, so necessarily $a_{n-1} = 0$.

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}(1+z-\sqrt{1-6z+z^2})$$

The general template is

$$p(z)S(z) + q(z) = \sqrt[k]{r(z)}$$

with $p(z)$, $q(z)$, and $r(z)$ depending on z. Differentiate both sides:

$$p'(z)S(z) + p(z)S'(z) + q'(z) = \frac{1}{k}r(z)^{\frac{1}{k}-1}r'(z)$$

Then multiply through by $r(z)$:

$$p'(z)r(z)S(z) + p(z)r(z)S'(z) + q'(z)r(z) = \frac{1}{k}(p(z)S(z)+q(z))r'(z)$$

After rearranging,

$$p(z)r(z)S'(z) + (p'(z)r(z) - \frac{1}{k}p(z)r'(z))S(z) = \frac{1}{k}q(z)r'(z) - q'(z)r(z)$$

Now plug in $p(z) = -4$, $q(z) = 1+z$, $k = 2$, and $r(z) = 1-6z+z^2$. A short simplification yields:

$$(z^2 - 6z + 1)S'(z) + (3 - z)S(z) = 1 - z$$

from which the recurrence follows.

▷ **I.43.** *Fast determination of the Cayley–Pólya numbers.*

$$H(z) = z \exp\left(H(z) + \frac{1}{2}H(z^2) + \cdots\right)$$

Step 1. Take the natural logarithm.

The logarithm brings the exponent down to a more manageable level:

$$\ln H(z) = \ln z + \sum_{k=1}^{\infty} \frac{1}{k} H(z^k)$$

Step 2. Differentiate with respect to z.

Differentiating gives:

$$\frac{H'(z)}{H(z)} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{1}{k} H'(z^k) \cdot kz^{k-1} = \frac{1}{z} + \sum_{k=1}^{\infty} z^{k-1} H'(z^k)$$

Step 3. Multiply by $zH(z)$.

Multiplying by $zH(z)$ clears the fractions and lines up the powers of z:

$$zH'(z) = H(z)\left(1 + \sum_{k=1}^{\infty} z^k H'(z^k)\right)$$

Step 4. Substitute power series.

Now expand $H(z)$ as a formal power series $\sum_{n=1}^{\infty} H_n z^n$. Then

- $zH'(z) = \sum_{n=1}^{\infty} nH_n z^n$
- $z^k H'(z^k) = \sum_{m=1}^{\infty} mH_m z^{mk}$

Substituting these series into the equation,

$$\sum_{n=1}^{\infty} nH_n z^n = H(z) + H(z) \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} mH_m z^{mk}$$

Step 5. The recurrence relation.

Extracting the coefficient of z^n leads to the recurrence. The double sum is a divisor-type convolution, and the standard form is

$$(n-1)H_n = \sum_{l=1}^{n-1} H_{n-l} \left(\sum_{d|l} dH_d\right)$$

or, solving explicitly for H_n,

$$H_n = \frac{1}{n-1}\sum_{l=1}^{n-1} H_{n-l}\left(\sum_{d|l} dH_d\right) = \frac{1}{n-1}\sum_{l=1}^{n-1} a_l H_{n-l}$$

To see the recurrence at work, start from the base case $H_1 = 1$, corresponding to the one-node tree.

- To find H_2, we set $n = 2$. The summation runs for $l = 1$. The only divisor of 1 is 1. $a_1 = \sum_{d|1} dH_d = 1 \cdot H_1 = 1 \implies H_2 = \frac{1}{2-1}(H_{2-1} \cdot a_1) = 1$.
- For H_3, the summation runs for $l = 1$ and $l = 2$. The divisors of 2 are 1 and 2. $a_2 = \sum_{d|2} dH_d = 1 \cdot H_1 + 2 \cdot H_2 = 3 \implies H_3 = \frac{1}{3-1}(H_{3-1} \cdot a_1 + H_{3-2} \cdot a_2) = 2$.
- For H_4, the summation runs for $l = 1, 2, 3$. The divisors of 3 are 1 and 3. $a_3 = \sum_{d|3} dH_d = 1 \cdot H_1 + 3 \cdot H_3 = 7 \implies H_4 = \frac{1}{4-1}(H_{4-1} \cdot a_1 + H_{4-2} \cdot a_2 + H_{4-3} \cdot a_3) = 4$.

▷ **I.50.** *Excursions, bridges, and meanders.*

A meander ending at altitude k can be read as k up-steps that create new floors, with Dyck excursions $\mathcal{D}$ inserted between consecutive floors and after the last one.

$$\mathcal{M} = \mathcal{D} + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + \cdots = \mathcal{D} \times \sum_{k=0}^{\infty} (\mathcal{Z} \times \mathcal{D})^k$$

A bridge is a sequence of primitive excursions. Each primitive excursion leaves 0, avoids returning in the middle, and comes back only at the end.

- **Going Above:** Starts with +1, ends with -1, with a Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$
- **Going Down:** Starts with -1, ends with 1, with a mirrored Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$

$$\mathcal{B} = \textsc{Seq}(\mathcal{Z} \times \mathcal{D} \times \mathcal{Z} + \mathcal{Z} \times \mathcal{D} \times \mathcal{Z}) = \textsc{Seq}(2 \times \mathcal{Z}^2 \times \mathcal{D})$$

To prove $M_n = \binom{n}{\lfloor n/2 \rfloor}$, let $N(n,k)$ be the number of unrestricted paths of length n ending at altitude k. The reflection principle does the main work: paths that ever hit -1 and end at $k \geq 0$ are in bijection with unrestricted paths ending at $-k-2$. So the number of meanders ending at k is $N(n,k) - N(n,k+2)$, and summing over $k \geq 0$ gives the telescoping identity

$$M_n = \sum_{k \geq 0} [N(n,k) - N(n,k+2)]$$

- n is even $(n = 2m)$ $M_n = N(n,0) = N(2m,0) = \binom{2m}{\frac{2m+0}{2}} = \binom{2m}{m} = \binom{n}{\lfloor n/2 \rfloor}$
- n is odd $(n = 2m+1)$ $M_n = N(n,1) = N(2m+1,1) = \binom{2m+1}{\frac{2m+1+1}{2}} = \binom{2m+1}{m} = \binom{n}{\lfloor n/2 \rfloor}$

***Example* I.17.** *The complexity of boolean functions.*

We claim that

$$D_n = O(16^n m^n n^{-3/2})$$

implies

$$\sum_{j=0}^{v} D_j = O(16^v m^v v^{-3/2})$$

The clean way is to prove a slightly more general statement first.
Let $c > 1$ and $\alpha > 0$. Define $a_n = c^n n^{-\alpha}$ for $n \geq 1$ and let $a_0 > 0$ be a constant. If $D_n = O(a_n)$, then $\sum_{j=0}^{v} D_j = O(c^v v^{-\alpha})$ as $v \to \infty$.

Proof. Since $D_n = O(a_n)$, there exist constants $M > 0$ and $N \geq 1$ such that $D_n \leq M c^n n^{-\alpha}$ for all $n \geq N$. For $v \geq N$ we split:

$$\sum_{j=0}^{v} D_j \leq \underbrace{\sum_{j=0}^{N-1} D_j}_{C_0} + M \sum_{j=N}^{v} c^j j^{-\alpha}$$

So the problem reduces to controlling

$$S(v) = \sum_{j=N}^{v} c^j j^{-\alpha}.$$

Split the sum at $j = \lfloor v/2 \rfloor$:

$$S(v) = \underbrace{\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha}}_{\text{early}} + \underbrace{\sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j j^{-\alpha}}_{\text{tail}}$$

Early terms. For $j \leq v/2$ we have $c^j \leq c^{v/2}$. Also since $\alpha > 0$, the factor $j^{-\alpha}$ is decreasing; its maximum is at $j = N$, so $j^{-\alpha} \leq N^{-\alpha}$. Thus

$$\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha} \leq N^{-\alpha} \sum_{j=N}^{\lfloor v/2 \rfloor} c^j \leq N^{-\alpha} \frac{c^{v/2+1}}{c-1} = O(c^{v/2})$$

Because $c > 1$, $c^{v/2}$ is exponentially smaller than c^v, so this part is negligible on the relevant scale.
Tail terms. For $j \geq v/2$, $j^{-\alpha} \leq (v/2)^{-\alpha} = 2^{\alpha} v^{-\alpha}$. Consequently

$$\sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j j^{-\alpha} \leq 2^{\alpha} v^{-\alpha} \sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j \leq 2^{\alpha} v^{-\alpha} \frac{c^{v+1}}{c-1} = O(c^v v^{-\alpha})$$

Putting the two pieces together, we get $S(v) = O(c^v v^{-\alpha})$, and hence

$$\sum_{j=0}^{v} D_j = C_0 + M S(v) = O(1) + O(c^v v^{-\alpha}) = O(c^v v^{-\alpha})$$

With the general estimate in hand, return to the original problem. We now show

$$\sum_{j=0}^{v} D_j = O(16^v m^v v^{-3/2}) = o(\|\Omega\|)$$

by taking for v the value

$$v = \frac{2^m}{4 + \log_2 m}$$

Observe that

$$v \log_2(16m) = \frac{2^m}{4 + \log_2 m}(4 + \log_2 m) = 2^m \implies 16^v m^v = 2^{2^m}$$

So the bound simplifies to

$$\sum_{j=0}^{v} D_j = O(2^{2^m} v^{-3/2})$$

Dividing by $\|\Omega\|$:

$$\frac{\sum_{j=0}^{v} D_j}{\|\Omega\|} = O(v^{-3/2})$$

As $m \to \infty$, we also have $v \to \infty$, hence $v^{-3/2} \to 0$. Therefore,

$$\frac{\sum_{j=0}^{v} D_j}{\|\Omega\|} \to 0$$

which is exactly $\sum_{j=0}^{v} D_j = o(\|\Omega\|)$.

1.6 Additional constructions

▷ **I.62.** *Sequences without repeated components.*

A sequence without repeated components is an ordered arrangement of distinct objects drawn from a base class $\mathcal{B}$. The natural route is to build the set first and recover the ordering afterward. The generating function for the power-set construction $\text{PSET}(\mathcal{B})$ is

$$A(z, u) = \prod_{\beta \in \mathcal{B}} \left(1 + u z^{|\beta|}\right) = \exp\left(\sum_{j \geq 1} (-1)^{j-1} \frac{u^j}{j} B(z^j)\right)$$

Here $[u^k]A(z, u) = f^{\langle k \rangle}(z)$ is the generating function for sets with exactly k distinct components. The no-repeat condition is already built in, but order is not. To turn sets into sequences, we multiply by $k!$ to account for all permutations. Using the Eulerian integral representation $k! = \int_0^\infty e^{-u} u^k \, du$, we can write the generating function for distinct sequences, $\mathcal{S}_{\text{distinct}}(z)$, as follows:

$$\begin{aligned}
\mathcal{S}_{\text{distinct}}(z) &= \sum_{k\geq 0} (\text{Sets of size k}) \cdot k! \\
&= \sum_{k\geq 0} f^{\langle k\rangle}(z) \cdot k! \\
&= \sum_{k\geq 0} f^{\langle k\rangle}(z) \cdot \int_0^\infty e^{-u} u^k \;\mathrm{d}u \\
&= \sum_{k\geq 0} \int_0^\infty f^{\langle k\rangle}(z) e^{-u} u^k \;\mathrm{d}u \\
&= \int_0^\infty \sum_{k\geq 0} f^{\langle k\rangle}(z) e^{-u} u^k \;\mathrm{d}u \\
&= \int_0^\infty \sum_{k\geq 0} f^{\langle k\rangle}(z) u^k \cdot e^{-u} \;\mathrm{d}u \\
&= \int_0^\infty A(z,u) \cdot e^{-u} \;\mathrm{d}u \\
&= \int_0^\infty \exp\left(\sum_{j\geq 1} (-1)^{j-1} \frac{u^j}{j} B(z^j) \right) e^{-u} \;\mathrm{d}u
\end{aligned}$$

1.7 Perspective

Chapter 2

LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

2.1 Labelled classes

2.2 Admissible labelled constructions

2.3 Surjections, set partitions, and words

2.4 Alignments, permutations, and related structures

2.5 Labelled trees, mappings, and graphs

▷ **II.19.** *Labelled hierarchies.*

We want to obtain

$$L(z) = T\left(\frac{1}{2}e^{z/2-1/2}\right) + \frac{z}{2} - \frac{1}{2}$$

from

$$L = z + e^L - 1 - L$$

Step 1. Re-arranging the target equation.

Collect the L-terms on one side:

$$2L + 1 - e^L = z$$

This still does not resemble $T(z) = ze^{T(z)}$, so we nudge it into Lambert-W territory.

Step 2. Utilizing the Lambert W identity.

The Cayley tree function is really Lambert W in disguise. In particular, $T(z) = ze^{T(z)}$ can be rewritten as

$$-T(z)e^{-T(z)} = -z \leftrightarrow z = -T(-z)e^{-T(-z)}$$

So $W(z) = -T(-z)$, equivalently $T(z) = -W(-z)$.

Step 3. Substitution and solving.

Return to the original equation and isolate the exponential:

$$e^L = 2L - z + 1$$

Whenever a variable appears both linearly and in the exponent, Lambert W is usually nearby. So rewrite the equation in the form $f(L)e^{f(L)} =$ something:

$$1 = (2L - z + 1)e^{-L}$$

To make the linear term match the exponent, set

$$u = \frac{z-1}{2} - L.$$

Then

$$2u = z - 1 - 2L = -(2L - z + 1)$$

Substituting this into the previous equation,

$$1 = (-2u)e^{-L}$$

Also, since $u = \frac{z-1}{2} - L$, we have $-L = u - \frac{z-1}{2}$. Plugging that into the exponent gives:

$$1 = -2u \cdot e^{u - \frac{z-1}{2}} = -2u \cdot e^u \cdot e^{-\frac{z-1}{2}}$$

Now isolate ue^u:

$$ue^u = -\frac{1}{2}e^{\frac{z-1}{2}}$$

Since $ue^u = v$ implies $u = W(v)$, and we know $W(v) = -T(-v)$:

$$u = W\left(-\frac{1}{2}e^{\frac{z-1}{2}}\right) = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

Substitute u back in:

$$\frac{z-1}{2} - L = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

and solve for L:

$$L = T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right) + \frac{z-1}{2}$$

II.5.2. *Mappings and associated graphs.*

Let $\mathcal{T}$ be the class of all Cayley trees. Then:

$$F = (1 - T)^{-1} \implies F_n = n^n$$

This follows neatly from Lagrange inversion. The standard form says

$$z = T/\phi(T) \implies [z^n]T(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n$$

More generally,

$$[z^n]H(T(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n$$

Taking $H(w) = (1-w)^{-1}$ and $\phi(w) = e^w$ gives:

$$\begin{aligned}
F_n &= n![z^n]F(z) \\
&= n!\frac{1}{n}[w^{n-1}]\left(\frac{1}{1-w}\right)'(e^w)^n \\
&= (n-1)![w^{n-1}]\frac{e^{nw}}{(1-w)^2} \\
&= (n-1)![w^{n-1}]\sum_{k=0}^{\infty}(k+1)w^k \cdot \sum_{j=0}^{\infty}\frac{n^j}{j!}w^j \\
&= (n-1)!\sum_{j=0}^{n-1}((n-1-j)+1)\frac{n^j}{j!} \\
&= (n-1)!\sum_{j=0}^{n-1}(n-j)\frac{n^j}{j!} \\
&= (n-1)!\left(\sum_{j=0}^{n-1}n\frac{n^j}{j!} - \sum_{j=0}^{n-1}j\frac{n^j}{j!}\right) \\
&= (n-1)!\left(\sum_{j=0}^{n-1}\frac{n^{j+1}}{j!} - \sum_{j=1}^{n-1}\frac{n^j}{(j-1)!}\right) \\
&= (n-1)!\left(\sum_{j=0}^{n-1}\frac{n^{j+1}}{j!} - \sum_{j=0}^{n-2}\frac{n^{j+1}}{j!}\right) \\
&= (n-1)!\frac{n^n}{(n-1)!} \\
&= n^n
\end{aligned}$$

Next consider $K = \log\left((1-T)^{-1}\right)$. Then

$$K_n = n^{n-1}Q(n)$$

where

$$Q_n = 1 + \frac{n-1}{n} + \frac{(n-1)(n-2)}{n^2} + \cdots$$

From Note II.18 we already know

$$n![z^n]\frac{T(z)^k}{k!} = \binom{n-1}{k-1}n^{n-k} \implies n![z^n]\frac{T(z)^k}{k} = (k-1)!\binom{n-1}{k-1}n^{n-k} = \frac{(n-1)!}{(n-k)!}n^{n-k}$$

Therefore,

$$\begin{aligned}
K_n &= n![z^n]K(z) \\
&= n![z^n]\sum_{j=1}^{\infty}\frac{T(z)^j}{j} \\
&= \sum_{j=1}^{\infty} n![z^n]\frac{T(z)^j}{j} \\
&= \sum_{j=1}^{\infty}\frac{(n-1)!}{(n-j)!}n^{n-j} \\
&= n^{n-1}\sum_{j=1}^{\infty}\frac{(n-1)!}{(n-j)!n^{j-1}} \\
&= n^{n-1}\left(1+\frac{n-1}{n}+\frac{(n-1)(n-2)}{n^2}+\cdots\right)
\end{aligned}$$

II.5.2. *Constrained mappings.*

The class of maps without fixed points has exponential generating function:

$$\frac{e^{-T(z)}}{1-T(z)}$$

and we want to show

$$n![z^n]\frac{e^{-T(z)}}{1-T(z)} = (n-1)^n$$

Again, Lagrange inversion does the bookkeeping. With $H(w) = \frac{e^{-w}}{1-w}$ and $\phi(w) = e^w$, we obtain

$$\begin{aligned}
n![z^n]\frac{e^{-T(z)}}{1-T(z)} &= n!\frac{1}{n}[w^{n-1}]\left(\frac{e^{-w}}{1-w}\right)'(e^w)^n \\
&= (n-1)![w^{n-1}]\frac{we^{(n-1)w}}{(1-w)^2} \\
&= (n-1)![w^{n-2}]\frac{e^{(n-1)w}}{(1-w)^2} \\
&= (n-1)![w^{n-2}]\sum_{k=0}^{\infty}(k+1)w^k\cdot\sum_{j=0}^{\infty}\frac{(n-1)^j}{j!}w^j \\
&= (n-1)!\sum_{j=0}^{n-2}((n-2-j)+1)\frac{(n-1)^j}{j!} \\
&= (n-1)!\sum_{j=0}^{m-1}(m-j)\frac{m^j}{j!} \\
&= (n-1)!\frac{m^m}{(m-1)!} \\
&= (n-1)!\frac{(n-1)^{n-1}}{(n-2)!} \\
&= (n-1)^n
\end{aligned}$$

II.5.3. ***Unrooted trees and acyclic graphs.***

To see why $U(z) = \int_0^z \frac{T(w)}{w}\,\mathrm{d}w$, start from the exponential generating functions

$$T(z) = \sum_{n\geq 0} T_n \frac{z^n}{n!}, \quad U(z) = \sum_{n\geq 0} U_n \frac{z^n}{n!}$$

Differentiate $U(z)$ and multiply by z; using $T_0 = 0$, this gives:

$$z\frac{\mathrm{d}}{\mathrm{d}z}U(z) = z\frac{\mathrm{d}}{\mathrm{d}z}\sum_{n\geq 0} U_n\frac{z^n}{n!} = z\sum_{n\geq 1} nU_n\frac{z^{n-1}}{n!} = \sum_{n\geq 1} T_n\frac{z^n}{n!} = T(z)$$

So $\frac{\mathrm{d}}{\mathrm{d}z}U(z) = \frac{T(z)}{z}$. Integrating from 0 to z gives:

$$U(z) - U(0) = \int_0^z \frac{T(w)}{w}\,\mathrm{d}w$$

Now use the functional equation for rooted trees:

$$T(z) = ze^{T(z)}$$

This is equivalent to $z = Te^{-T}$. Differentiate with respect to T:

$$\mathrm{d}z = \left(e^{-T} - Te^{-T}\right)\mathrm{d}T = e^{-T}(1-T)\,\mathrm{d}T$$

Substituting $z = Te^{-T}$ and $\mathrm{d}z$ into the integral turns it into

$$U(z) = \int \frac{T}{Te^{-T}} \cdot e^{-T}(1-T)\ \mathrm{d}T = \int (1-T)\ \mathrm{d}T = T - \frac{T^2}{2} + C$$

Since $U(0) = 0$ and $T(0) = 0$, the constant is $C = 0$.

▷ **II.22.** *2–Regular graphs.*

The clean way to view this is as a bijection between clouds and 2-regular graphs on n labelled vertices. Number the lines $1, 2, \ldots, n$.

- Line i in the cloud $\leftrightarrow$ vertex i in the graph.
- Intersection point $\{i,\ j\}$ in the cloud $\leftrightarrow$ edge $i - j$ in the graph.

Let n_i be the number of chosen intersection points on line i. Since each point lies on two lines, $\sum_{i=1}^n n_i = 2n$. *If no three points are collinear*, then $n_i \leq 2$ for all i, so $\sum_{i=1}^n n_i \leq 2n$. Equality $\sum_{i=1}^n n_i = 2n$ forces $n_i = 2$ for every i, so every vertex has degree 2.

Conversely, if every vertex has degree 2, then $n_i = 2$ for all i, so each line contains exactly two intersection points; in particular, no three chosen points are collinear.

So the two conditions are equivalent.

▷ **II.23.** *Wright's surgery.*

Step 1. Deconstruct the graph.

Start with a connected graph G of excess $k+1$. Choose an edge and orient it, so we have a source vertex A and a target vertex B. Remove that directed edge. At that point exactly two things can happen:

- **Case 1 (Connected)**: The removed edge lay on a cycle, so the graph stays connected. We lost one edge and no vertices, hence the excess drops to k. What remains is a connected graph with two ordered distinguished vertices, A and B.
- **Case 2 (Disconnected)**: The removed edge was a bridge, so the graph splits into two connected components G_1 and G_2. If G_1 has excess h, then G_2 has excess $k-h$. Each component carries one distinguished vertex, recording where the deleted edge used to attach.

Step 2. Reconstruct the graph.

To reconstruct all directed-edge-pointed graphs of excess $k+1$, reverse the two cases:

- **Reversing Case 1**: Start from one connected graph of excess k. Choose an ordered pair of distinct vertices A, B and draw a directed edge from A to B. If A and B were already adjacent, this would create a forbidden multi-edge, so those choices must be excluded.
- **Reversing Case 2**: Start from two independent connected components, of excess h and $k-h$. Choose one vertex in each and join them by a directed edge from A to B. This merges the two components into a connected graph of excess $k+1$.

Now match these operations with the differential recurrence:

$$2\partial_y w_{k+1} = \left(z^2\partial_z^2 w_k - 2y\partial_y w_k\right) + \sum_{h=-1}^{k+1} (z\partial_z w_h) \cdot (z\partial_z w_{k-h})$$

Step 3. Interpret the left-hand side.

- w_{k+1}: Represents the universe of connected graphs of excess $k+1$.
- ∂_y: Differentiation with respect to the edge-marker y means “choose an edge and delete it”.
- 2: The deleted edge is oriented, so we must decide which endpoint is the source and which is the target.
- Together, $2\partial_y w_{k+1}$ counts connected excess-$(k+1)$ graphs with one remembered directed edge removed.

Step 4. Interpret the connected term.

- w_k: The base graph of excess k.
- ∂_z^2: Differentiate twice with respect to the vertex-marker to select an ordered pair of distinct vertices A, B.
- z^2: Put the two marked vertices back into the count.
- $-2y\partial_y w_k$: This removes the bad choices where A and B were already joined by an edge. The factor 2 again accounts for orientation.

Step 5. Interpret the disconnected term.

- w_h and w_{k-h}: These are the two independent connected components, so their generating functions multiply.
- $z\partial_z$: This is the standard pointing operator for vertices.
  - Applied to the first component, it selects the source vertex A.
  - Applied to the second component, it selects the target vertex B.
- $\sum_{h=-1}^{k+1}$: Sum over all ways to distribute the remaining excess between the two components. The lower bound -1 appears because a tree already has excess -1.

2.6 Additional constructions

▷ **II.28.** *Do two permutations generate the symmetric group?*

We want to show

$$n!^2\pi_n = (n-1)!I_{n+1}$$

The first useful observation is that the exponential generating function of pairs of permutations,

$$\sum_{n\geq 0} n!^2\frac{z^n}{n!} = \sum_{n\geq 0} n!z^n,$$

is exactly the ordinary generating function $P(z)$ of permutations. By the exponential formula, the exponential generating function $C(z)$ of connected graphs therefore satisfies

$$C(z) = \log P(z) = \log\left(\sum_{n\geq 0} n!z^n\right)$$

Since every permutation decomposes uniquely into a sequence of indecomposable blocks, the generating function $I(z)$ of indecomposable permutations satisfies

$$I(z) = 1 - \frac{1}{P(z)}$$

Now differentiate and extract the consequence:

$$C'(z) = \frac{P'(z)}{P(z)} = P'(z)(1-I(z)) = P'(z) - P'(z)I(z)$$

Next we extract the coefficient of z^{n-1} from both sides.

Step 1. Extract the left-hand side coefficient.

Since $C(z) = \sum_{n\geq 1}(n!^2\pi_n)\frac{z^n}{n!}$, its derivative is:

$$C'(z) = \sum_{n\geq 1}(n!^2\pi_n)\frac{z^{n-1}}{(n-1)!}$$

So the coefficient of z^{n-1} is:

$$[z^{n-1}]C'(z) = \frac{n!^2\pi_n}{(n-1)!} = n \cdot n!\pi_n$$

Step 2. Extract the right-hand side coefficient.

Differentiate both sides of $P(z)(1 - I(z)) = 1$ with respect to z:

$$P'(z)(1 - I(z)) - P(z)I'(z) = 0 \implies P'(z) - P'(z)I(z) = P(z)I'(z)$$

So it remains to extract the coefficient of z^{n-1} from $P(z)I'(z)$:

$$[z^{n-1}](P'(z) - P'(z)I(z)) = [z^{n-1}]P(z)I'(z) = \sum_{k=1}^{n} kI_k(n-k)!$$

Call this sum S_n. The whole problem is now to show $S_n = I_{n+1}$. Extracting the coefficient of z^N from $P(z) = 1 + I(z)P(z)$ for $N \geq 1$ gives the recurrence

$$\sum_{k=1}^{N} I_k(N-k)! = N! \tag{2.1}$$

Apply 2.1 with $N = n + 1$:

$$\sum_{k=1}^{n+1} I_k(n+1-k)! = (n+1)! \implies \sum_{k=1}^{n} I_k(n+1-k)! = (n+1)! - I_{n+1} \tag{2.2}$$

Expand the left-hand side of 2.2:

$$\begin{aligned}\sum_{k=1}^{n} I_k(n+1-k)! &= \sum_{k=1}^{n} I_k(n+1-k)(n-k)! \\ &= (n+1)\sum_{k=1}^{n} I_k(n-k)! - \sum_{k=1}^{n} kI_k(n-k)!\end{aligned}$$

By 2.1 with $N = n$:

$$\sum_{k=1}^{n} I_k(n-k)! = n!$$

Hence the left-hand side of 2.2 is:

$$\sum_{k=1}^{n} I_k(n+1-k)! = (n+1)n! - S_n = (n+1)! - S_n$$

Step 3. Conclude the identity.

Substituting back into 2.2 gives $S_n = I_{n+1}$.

Finally, $n \cdot n!\pi_n = S_n$, so

$$n!^2\pi_n = (n-1)!I_{n+1}.$$

2.7 Perspective

Chapter 3

COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

3.1 An introduction to bivariate generating functions (BGFs)

3.2 Bivariate generating functions and probability distributions

3.3 Inherited parameters and ordinary MGFs

$\triangleright$ **III.7.** *Largest summands in compositions.*

Let M be the largest summand in a random composition. We claim that for any $\epsilon \in (0,1)$,

$$\mathbb{P}\left((1-\epsilon)\log_2 n \leq M \leq (1+\epsilon)\log_2 n\right) \to 1$$

Step 1. Upper bound via the first moment method.

Set $r_+ = \lfloor (1+\epsilon)\log_2 n \rfloor$. If $M \geq r_+$, then some summand has size at least r_+. Let $S = \sum_{k=r_+}^{\infty} X_k$, where X_k counts summands of size exactly k. Then Markov gives:

$$\mathbb{P}(M \geq r_+) = \mathbb{P}(S \geq 1) \leq \frac{\mathbb{E}[S]}{1} = \sum_{k=r_+}^{\infty} \mathbb{E}[X_k]$$

Proposition III.4 gives $\mathbb{E}[X_k] = \frac{n-k+3}{2^{k+1}} \leq \frac{n}{2^k}$. Hence, noting that $X_k = 0$ for $k > n$,

$$\mathbb{E}[S] \leq \sum_{k=r_+}^{n} \frac{n}{2^k} < \frac{n}{2^{r_+-1}} < \frac{n}{2^{(1+\epsilon)\log_2 n - 2}} = 4n^{-\epsilon} \to 0$$

Since $M > (1+\epsilon)\log_2 n$ implies $M \geq \lfloor (1+\epsilon)\log_2 n \rfloor = r_+$, we have $\mathbb{P}(M > (1+\epsilon)\log_2 n) \leq \mathbb{P}(M \geq r_+) \to 0$.

Step 2. Lower bound via the second moment method.

For the lower bound, view a composition of n as $n-1$ independent gaps, each cut with probability 1/2. Let $r_- = \lceil (1-\epsilon)\log_2 n \rceil$. Split the n ones into $m = \left\lfloor \frac{n}{r_-} \right\rfloor$ disjoint blocks of length r_-, ignoring the leftovers. Each block has $r_- - 1$ internal gaps, so it contains no internal cut with probability $p = \left(\frac{1}{2}\right)^{r_- - 1}$.

Define the indicator random variables

$$I_i = \begin{cases} 1 & \text{if block i has no internal cuts} \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, \cdots, m$. Let $Y = \sum_{i=1}^{m} I_i$, the number of blocks that survive intact. If $Y \geq 1$, then at least one block forms a summand of size at least r_-, so $\mathbb{P}(M \geq r_-) \geq \mathbb{P}(Y > 0)$.

Now the Paley–Zygmund inequality does exactly the required job:

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

In our setting, $Y \sim \text{Binomial}(m, p)$.

First compute the expectation:

$$\mathbb{E}[Y] = mp = \left\lfloor \frac{n}{r_-} \right\rfloor \left(\frac{1}{2}\right)^{r_- - 1}$$

Estimate the two factors separately:

- For the floor term $\lfloor n/r_- \rfloor$:

$$\left\lfloor \frac{n}{r_-} \right\rfloor > \frac{n}{r_-} - 1 = \frac{n}{\lceil (1-\epsilon)\log_2 n \rceil} - 1 > \frac{n}{(1-\epsilon)\log_2 n + 1} - 1$$

  For sufficiently large n, this can be further simplified to:

$$\frac{n}{(1-\epsilon)\log_2 n + 1} - 1 > \frac{n}{2(1-\epsilon)\log_2 n} - 1 > \frac{n}{4(1-\epsilon)\log_2 n}$$

- For the exponential term $(1/2)^{r_- - 1}$:

$$\left(\frac{1}{2}\right)^{r_- - 1} > \left(\frac{1}{2}\right)^{(1-\epsilon)\log_2 n} = n^{-(1-\epsilon)}$$

Putting these together,

$$\mathbb{E}[Y] > \frac{n^{\epsilon}}{4(1-\epsilon)\log_2 n} \to \infty \quad \text{as } n \to \infty$$

Next control the second moment. Since Y is binomial,

$$\text{Var}(Y) = mp(1-p) \leq mp = \mathbb{E}[Y]$$

Substitute this into Paley–Zygmund:

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]} = \frac{(\mathbb{E}[Y])^2}{\text{Var}(Y) + (\mathbb{E}[Y])^2} \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y] + (\mathbb{E}[Y])^2} = \frac{1}{1 + \frac{1}{\mathbb{E}[Y]}} \to 1$$

Finally, $M \geq r_-$ implies $M \geq (1-\epsilon)\log_2 n$, so

$$\mathbb{P}(M \geq (1-\epsilon)\log_2 n) \geq \mathbb{P}(M \geq r_-) \geq \mathbb{P}(Y > 0) \to 1$$

Step 3. Direct proof of the Paley–Zygmund bound.

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

Start from $\mathbb{E}[Y]$ and split according to whether $Y > 0$:

$$\mathbb{E}[Y] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}] + \mathbb{E}[Y \cdot \mathbf{1}_{Y=0}] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}]$$

Now apply Cauchy–Schwarz:

$$\mathbb{E}[Y] \leq \sqrt{\mathbb{E}[Y^2]}\sqrt{\mathbb{E}[\mathbf{1}_{Y>0}]} = \sqrt{\mathbb{E}[Y^2]}\sqrt{\mathbb{P}(Y > 0)}$$

Since everything is nonnegative, we may square both sides:

$$(\mathbb{E}[Y])^2 \leq \mathbb{E}[Y^2] \cdot \mathbb{P}(Y > 0) \implies \mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

3.4 Inherited parameters and exponential MGFs

$\triangleright$ **III.10.** *A hundred prisoners II.*

The randomisation trick is easiest to see on a small example. Suppose there are 10 drawers, and the director arranges them in one giant cycle $1 \to 2 \to 3 \to \cdots \to 10 \to 1$. Then any prisoner who follows the cycle strategy but may open only 5 drawers is doomed: prisoner 1 will not see "1" until the 10th step.

Before the game, the prisoners agree on a random relabelling. Say their secret permutation is: "Drawer 1 means 7, 2 means 3, 3 means 9, 4 means 1, 5 means 8, 6 means 2, 7 means 10, 8 means 4, 9 means 6, 10 means 5." Now when a prisoner says "open drawer n", they actually open the physical drawer assigned to n by this secret code. Under that relabelling, the same deterministic strategy behaves quite differently:

- Prisoner 1
  - Starts at "Drawer 1" $\to$ goes to Physical Drawer 7.
  - Finds Card 8.
  - Goes to "Drawer 8" $\to$ goes to Physical Drawer 4.
  - Finds Card 5.
  - Goes to "Drawer 5" $\to$ goes to Physical Drawer 8.
  - Finds Card 9.
  - Goes to "Drawer 9" $\to$ goes to Physical Drawer 6.
  - Finds Card 7.
  - Goes to "Drawer 7" $\to$ goes to Physical Drawer 10.
  - Finds Card 1. Success! Prisoner 1 found his number in 5 steps.

- Prisoner 2
  - Starts at "Drawer 2" $\rightarrow$ goes to Physical Drawer 3.
  - Finds Card 4.
  - Goes to "Drawer 4" $\rightarrow$ goes to Physical Drawer 1.
  - Finds Card 2. Success! Prisoner 2 found his number in 2 steps.
- Prisoner 3
  - Starts at "Drawer 3" $\rightarrow$ goes to Physical Drawer 9.
  - Finds Card 10.
  - Goes to "Drawer 10" $\rightarrow$ goes to Physical Drawer 5.
  - Finds Card 6.
  - Goes to "Drawer 6" $\rightarrow$ goes to Physical Drawer 2.
  - Finds Card 3. Success! Prisoner 3 found his number in 3 steps.
- Prisoner 4
  - Starts at "Drawer 4" $\rightarrow$ goes to Physical Drawer 1.
  - Finds Card 2.
  - Goes to "Drawer 2" $\rightarrow$ goes to Physical Drawer 3.
  - Finds Card 4. Success! Prisoner 4 found his number in 2 steps.
- Prisoner 5
  - Starts at "Drawer 5" $\rightarrow$ goes to Physical Drawer 8.
  - Finds Card 9.
  - Goes to "Drawer 9" $\rightarrow$ goes to Physical Drawer 6.
  - Finds Card 7.
  - Goes to "Drawer 7" $\rightarrow$ goes to Physical Drawer 10.
  - Finds Card 1.
  - Goes to "Drawer 1" $\rightarrow$ goes to Physical Drawer 7.
  - Finds Card 8.
  - Goes to "Drawer 8" $\rightarrow$ goes to Physical Drawer 4.
  - Finds Card 5. Success! Prisoner 5 found his number in 5 steps.

$\triangleright$ **III.15.** *Distinct component sizes in set partitions.*

- Number of distinct block sizes in set partitions:

$$\prod_{n=1}^{\infty} \mathcal{E} + u\textsc{Set}_{\geq 1}(\textsc{Set}_n(\mathcal{Z}))$$

- Number of distinct cycle sizes in permutations:

$$\prod_{n=1}^{\infty} \mathcal{E} + u\textsc{Set}_{\geq 1}(\textsc{Cyc}_n(\mathcal{Z}))$$

The point in both cases is the same: for each admissible size n, we either use no component of size n, or we use a non-empty set of them and mark that size once with u.

3.5 Recursive parameters

***Example* III.14.** *Leaves and node types in binary trees.*

Strictly speaking, the relevant object here is not the full class of binary trees but the class $\bar{\mathcal{B}}$ of *pruned binary trees* (see ***Example* I.13.**). So we begin with the correct class:

$$\hat{B}(z) = C(z) - 1 = \frac{1-\sqrt{1-4z}}{2z} - 1 = \frac{1-2z-\sqrt{1-4z}}{2z}$$

where $C(z)$ is the usual Catalan generating function, satisfying $C(z) = 1 + zC(z)^2$.

Leaves

$$\hat{B}(z,u) = zu + 2z\hat{B}(z,u) + z\hat{B}(z,u)^2 \implies \frac{\partial \hat{B}}{\partial u} = z + 2z\frac{\partial \hat{B}}{\partial u} + 2z\hat{B}\frac{\partial \hat{B}}{\partial u} \implies$$

$$\frac{\partial \hat{B}}{\partial u} = \frac{z}{1-2z-2z\hat{B}} \implies \left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = \frac{z}{1-2z-2z\hat{B}(z)} = \frac{z}{\sqrt{1-4z}} \implies$$

$$[z^n]\left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = [z^{n-1}](1-4z)^{-1/2} = \binom{2n-2}{n-1} = \frac{(2n-2)!}{(n-1)!^2}$$

1-nodes

$$\hat{B}(z,u) = z + 2zu\hat{B}(z,u) + z\hat{B}(z,u)^2 \implies \frac{\partial \hat{B}}{\partial u} = 2z\hat{B} + 2zu\frac{\partial \hat{B}}{\partial u} + 2z\hat{B}\frac{\partial \hat{B}}{\partial u} \implies$$

$$\frac{\partial \hat{B}}{\partial u} = \frac{2z\hat{B}}{1-2zu-2z\hat{B}} \implies \left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = \frac{2z\hat{B}(z)}{1-2z-2z\hat{B}(z)} = \frac{1-2z}{\sqrt{1-4z}} - 1 \implies$$

$$[z^n]\left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = [z^n]\left((1-4z)^{-1/2} - 2\frac{z}{\sqrt{1-4z}}\right) = \binom{2n}{n} - 2\binom{2n-2}{n-1}$$

2-nodes

$$\hat{B}(z,u) = z + 2z\hat{B}(z,u) + zu\hat{B}(z,u)^2 \implies \frac{\partial \hat{B}}{\partial u} = 2z\frac{\partial \hat{B}}{\partial u} + z\hat{B}^2 + 2zu\hat{B}\frac{\partial \hat{B}}{\partial u} \implies$$

$$\frac{\partial \hat{B}}{\partial u} = \frac{z\hat{B}^2}{1-2z-2zu\hat{B}} \implies \left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = \frac{z\hat{B}(z)^2}{1-2z-2z\hat{B}(z)} = \frac{z\hat{B}(z)^2}{\sqrt{1-4z}} = \frac{z^3C(z)^4}{\sqrt{1-4z}}$$

At this point a standard Catalan identity saves a fair amount of algebra. Since $C'(z)$ satisfies $C'(z) = C(z)^2 + 2zC(z)C'(z)$, we have

$$C'(z) = \frac{C(z)^2}{1-2zC(z)} = \frac{C(z)^2}{\sqrt{1-4z}}$$

Plugging that back in gives:

$$\left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} = z^3C(z)^2 \cdot \frac{C(z)^2}{\sqrt{1-4z}} = z^3C(z)^2C'(z) = \frac{z^3}{3}\frac{\mathrm{d}}{\mathrm{d}z}C(z)^3 = z^3\frac{\mathrm{d}}{\mathrm{d}z}F(z)$$

where $F(z) = \frac{1}{3}C(z)^3$. Now:

$$\begin{aligned}
[z^n]\left.\frac{\partial \hat{B}}{\partial u}\right|_{u=1} &= [z^n]z^3\frac{\mathrm{d}}{\mathrm{d}z}F(z) \\
&= [z^{n-3}]\frac{\mathrm{d}}{\mathrm{d}z}F(z) \\
&= (n-2)F_{n-2} \\
&= \frac{n-2}{3}[z^{n-2}]C(z)^3 \\
&= \frac{n-2}{3}\cdot\frac{3}{n+1}\binom{2n-2}{n-2} \\
&= \binom{2n-2}{n-3}
\end{aligned}$$

After collecting the three contributions, the total number of **Leaves** + **1-nodes** + **2-nodes** is

$$\binom{2n-2}{n-1} + \binom{2n}{n} - 2\binom{2n-2}{n-1} + \binom{2n-2}{n-3} = \frac{n}{n+1}\binom{2n}{n}$$

which is exactly what it should be.

▷ **III.18.** *Node-degree profile in simple varieties of trees.*

$$T(z,u) = z\left(\phi(T(z,u)) + \phi_k(u-1)T(z,u)^k\right)$$

Differentiating with respect to u gives:

$$\frac{\partial T}{\partial u} = z\left(\phi'(T)\frac{\partial T}{\partial u} + \phi_k T^k + k\phi_k(u-1)T^{k-1}\frac{\partial T}{\partial u}\right) \implies$$

$$\Omega(z) = z\left(\phi'(T(z))\Omega(z) + \phi_k T(z)^k\right) \implies \Omega(z) = z\frac{\phi_k T(z)^k}{1-z\phi'(T(z))}$$

Since $T(z) = z\phi(T(z))$, differentiating with respect to z yields:

$$T'(z) = \phi(T(z)) + z\phi'(T(z))T'(z) \implies 1 = \frac{\phi(T(z))}{T'(z)} + z\phi'(T(z))$$

Hence,

$$\Omega(z) = \phi_k z T(z)^k \cdot \frac{T'(z)}{\phi(T(z))} = \phi_k z T(z)^k \cdot \frac{T'(z)}{T(z)/z} = \phi_k z^2 T(z)^{k-1} T'(z)$$

For Cayley trees:

$$\begin{aligned}
n![z^n]\Omega(z) &= n![z^n]\frac{1}{k!}z^2\frac{\mathrm{d}}{\mathrm{d}z}\frac{T(z)^k}{k} \\
&= \frac{n!}{k \cdot k!}[z^{n-2}]\frac{\mathrm{d}}{\mathrm{d}z}T(z)^k \\
&= \frac{n!}{k \cdot k!}(n-1)[z^{n-1}]T(z)^k \\
&= \frac{n(n-1)}{k} \cdot (n-1)![z^{n-1}]\frac{T(z)^k}{k!} \\
&= \frac{n(n-1)}{k}\binom{n-2}{k-1}(n-1)^{n-1-k} \\
&= (n-k)\binom{n}{k}(n-1)^{n-1-k}
\end{aligned}$$

***Example* III.15.** *Path length in trees.*

We want to show

$$z^2\frac{G'(z)}{(1-G(z))^2 - z} = \frac{z}{2(1-4z)} - \frac{z}{2\sqrt{1-4z}}$$

The clean route is to rewrite everything in terms of G:

$$G = \frac{z}{1-G} \implies z = G(1-G) = G - G^2$$

$$z = G - G^2 \implies 1 = G' - 2GG' \implies G' = \frac{1}{1-2G}$$

Hence,

$$\begin{aligned}
z^2\frac{G'(z)}{(1-G(z))^2 - z} &= (G(1-G))^2\frac{1/(1-2G)}{1-2G+G^2-(G-G^2)} \\
&= \frac{G^2(1-G)^2}{(1-2G)(1-3G+2G^2)} \\
&= \frac{G^2(1-G)}{(1-2G)^2} \\
&= \frac{(G-G^2)\cdot G}{1-4(G-G^2)} \\
&= \frac{zG}{1-4z} \\
&= \frac{z}{1-4z}\cdot\frac{1-\sqrt{1-4z}}{2} \\
&= \frac{z}{2}\left(\frac{1}{1-4z} - \frac{1}{\sqrt{1-4z}}\right)
\end{aligned}$$

▷ **III.20.** *Path length in simple varieties of trees.*

Start from the functional equations

$$T(z) = z\phi(T(z)) \text{ where } T(z) = T(z,1); \quad T(z,u) = z\phi(T(zu,u))$$

Differentiate the second equation with respect to u:

$$\frac{\partial}{\partial u}T(z,u) = z\phi'(T(zu,u)) \cdot \frac{\mathrm{d}}{\mathrm{d}u}T(zu,u) = z\phi'(T(zu,u))[T_z(zu,u)z + T_u(zu,u)]$$

At $u = 1$, this becomes

$$T_u(z,1) = z\phi'(T(z))[zT'(z) + T_u(z,1)] \implies T_u(z,1) = \frac{z^2\phi'(T(z))T'(z)}{1 - z\phi'(T(z))} \tag{3.1}$$

Differentiating the first equation with respect to z gives:

$$T'(z) = \phi(T(z)) + z\phi'(T(z))T'(z) \implies \frac{1}{1 - z\phi'(T(z))} = \frac{T'(z)}{\phi(T(z))}$$

Insert this into 3.1:

$$T_u(z,1) = z^2\phi'(T(z))T'(z) \cdot \frac{T'(z)}{\phi(T(z))} = \frac{\phi'(T(z))}{\phi(T(z))}(zT'(z))^2$$

3.6 Complete generating functions and discrete models

***Example* III.16.** *Words and records.*

$$\begin{aligned}
[z^n]\frac{z}{1-rz}\sum_{j=1}^{r}\frac{1}{1-(j-1)z} &= [z^n]\frac{z}{1-rz}\sum_{k=0}^{r-1}\frac{1}{1-kz} \\
&= [z^n]\sum_{k=0}^{r-1}\frac{z}{(1-rz)(1-kz)} \\
&= [z^n]\sum_{k=0}^{r-1}\frac{1}{r-k}\left(\frac{rz}{1-rz} - \frac{kz}{1-kz}\right) \\
&= \sum_{k=0}^{r-1}\frac{1}{r-k}\left([z^n]\frac{rz}{1-rz} - [z^n]\frac{kz}{1-kz}\right) \\
&= \sum_{k=0}^{r-1}\frac{r^n - k^n}{r-k}
\end{aligned}$$

$$\mathbb{E}_{\mathcal{W}_n}(\#\text{records}) = \sum_{k=0}^{r-1}\frac{1-(k/r)^n}{r-k} = \sum_{k=0}^{r-1}\frac{1}{r-k} - \sum_{k=0}^{r-1}\frac{(k/r)^n}{r-k} = H_r - \sum_{k=1}^{r-1}\frac{(k/r)^n}{r-k}$$

***Example* III.19.** *Rises in Bernoulli trials: Simon Newcomb's problem.*

Step 1. Pass from W_j to X_j.

Start from the recurrence:

$$W_j = z_j + z_j \sum_{k=1}^{j-1} W_k + v z_j \sum_{k=j}^{r} W_k, \tag{3.2}$$

It is convenient to pass to X_j:

$$X_j = 1 + \sum_{k=1}^{j-1} W_k + v \sum_{k=j}^{r} W_k = 1 + \sum_{k=1}^{j-1} z_k X_k + v \sum_{k=j}^{r} z_k X_k. \tag{3.3}$$

Now take the forward difference:

$$X_{j+1} - X_j = W_j - vW_j = z_j X_j (1 - v),$$

This gives the first-order recurrence:

$$X_{j+1} = X_j(1 + z_j(1 - v)). \tag{3.4}$$

Step 2. Solve the first-order recurrence.

Unwinding (3.4) shows that every X_j is determined by X_1:

$$X_j = X_1 \prod_{k=1}^{j-1} (1 + (1 - v) z_k). \tag{3.5}$$

Step 3. Determine X_1.

To determine X_1, look at the boundary case $j = 1$:

$$W_1 = z_1 + v z_1 \sum_{k=1}^{r} W_k \implies z_1 X_1 = z_1 + v z_1 \sum_{k=1}^{r} z_k X_k.$$

Canceling z_1 gives:

$$X_1 = 1 + v \sum_{k=1}^{r} z_k X_k.$$

To evaluate the remaining sum, introduce the formal boundary term X_{r+1} by setting $j = r + 1$ in (3.3). Then the second sum disappears, leaving

$$X_{r+1} = 1 + \sum_{k=1}^{r} z_k X_k \implies \sum_{k=1}^{r} z_k X_k = X_{r+1} - 1.$$

Substituting back gives:

$$X_1 = 1 + v(X_{r+1} - 1).$$

On the other hand, (3.5) at $j = r + 1$ gives $X_{r+1} = X_1 P$, where $P = \prod_{j=1}^{r} (1 + (1 - v) z_j)$. Plugging this into the equation for X_1,

$$X_1 = 1 + v(X_1 P - 1),$$

and solving for X_1 gives:

$$X_1 = \frac{1 - v}{1 - vP}. \tag{3.6}$$

Step 4. Recover the total generating function.

Finally, the total generating function is:

$$W = 1 + \sum_{k=1}^{r} W_k = 1 + \sum_{k=1}^{r} z_k X_k = X_{r+1} = \frac{P(1 - v)}{1 - vP}. \tag{3.7}$$

3.7 Additional constructions

***Example* III.22.** *Constrained integer compositions and "slicing".*

This example concerns two independent regions, $\mathcal{R}_1$ and $\mathcal{R}_2$. They illustrate the same slicing philosophy, but the functional equations and resulting series are derived separately.

Region $\mathcal{R}_1$

Step 1. Derive the functional equation for $\mathcal{R}_1$.

$$\mathcal{R}_1 = \{(x, y)|1 \leq x \leq y\}$$

To obtain the functional equation, note that

$$\begin{aligned}
f(z) &= z + z^2 + \cdots = \frac{z}{1-z} \\
\mathcal{L}[F(z,u)] &= \mathcal{L}[\sum_j f^{\langle j\rangle}(z)u^j] \\
&= \sum_j f^{\langle j\rangle}(z)\mathcal{L}[u^j] \\
&= \sum_j f^{\langle j\rangle}(z)\frac{u^j}{1-u} \\
&= \frac{1}{1-u}F(z,u)
\end{aligned}$$

Therefore,

$$F(z,u) = f(zu) + (\mathcal{L}[F(z,u)])_{u\mapsto zu} = \frac{zu}{1-zu} + \frac{1}{1-zu}F(z,zu) \tag{3.8}$$

Step 2. Check the closed form for $\mathcal{R}_1$.

Now let us check that the two advertised forms of $F(z, u)$ really agree:

$$F(z,u) = \frac{zu}{1-zu} + \frac{z^2u}{(1-zu)(1-z^2u)} + \frac{z^3u}{(1-zu)(1-z^2u)(1-z^3u)} + \cdots \tag{3.9}$$

$$F(z,u) = \frac{zu}{1-z} + \frac{z^2u^2}{(1-z)(1-z^2)} + \frac{z^3u^3}{(1-z)(1-z^2)(1-z^3)} + \cdots \tag{3.10}$$

3.9 comes from repeated substitution. One could verify 3.10 by series expansion, but there is a cleaner algebraic route. Assume $F(z, u) = \sum_{k\geq 1} A_k(z)u^k$ and substitute into the functional equation (3.8):

$$(1 - zu)F(z, u) = zu + F(z, zu)$$

Expanding both sides gives:

$$\sum_{k\geq 1} A_k(z)u^k - zu\left(\sum_{k\geq 1} A_k(z)u^k\right) = zu + \sum_{k\geq 1} A_k(z)(zu)^k$$

Now collect powers of u:

$$\sum_{k\geq 1} A_k(z)u^k - \sum_{k\geq 1} A_k(z)zu^{k+1} = zu + \sum_{k\geq 1} A_k(z)z^k u^k$$

Extract the coefficients of u^k (for $k > 1$):

$$A_k(z) - zA_{k-1}(z) = z^k A_k(z) \implies A_k(z) = \frac{z}{1-z^k}A_{k-1}(z) \tag{3.11}$$

Starting from $A_1(z) = z + z^2 + \cdots = \frac{z}{1-z}$, the recurrence telescopes to

$$A_k(z) = \frac{z}{1-z^k}\frac{z}{1-z^{k-1}}\cdots\frac{z}{1-z^2}\frac{z}{1-z} = \frac{z^k}{(1-z)(1-z^2)\cdots(1-z^k)}$$

So the kth term of 3.10 is exactly $A_k(z)u^k$.

Step 3. Compare the two combinatorial interpretations.

There is also a nice combinatorial punchline: the kth term of 3.9 counts partitions with exactly k summands, whereas the kth term of 3.10 counts partitions whose largest summand is exactly k. The case $k = 3$ makes the contrast visible.

The 3rd term in 3.9:

$$\frac{z^3u}{(1-zu)(1-z^2u)(1-z^3u)} = (zzzu)\sum_j (zu)^j \sum_k z^k(zu)^k \sum_l z^l z^l (zu)^l$$

- $z^1 \cdot z^1 \cdot (zu)^1$ places 1 cell on all three columns
- $z^0 \cdot z^0 \cdot (zu)^j$ places $j \geq 0$ cells on the rightmost column
- $z^0 \cdot z^k \cdot (zu)^k$ places $k \geq 0$ cells on the rightmost two columns
- $z^l \cdot z^l \cdot (zu)^l$ places $l \geq 0$ cells on all three columns

This lifting mechanism is what keeps the summands weakly increasing.

The 3rd term in 3.10:

$$\frac{z^3u^3}{(1-z)(1-z^2)(1-z^3)}$$

- z^3u^3 ensures there is at least one part of size 3 (which also marks the last column height 3).
- $\frac{1}{(1-z)(1-z^2)(1-z^3)}$ is the standard generating function for partitions using parts of size $\{1, 2, 3\}$.

We now leave $\mathcal{R}_1$ entirely and start a separate analysis for $\mathcal{R}_2$.

Region $\mathcal{R}_2$

Step 1. Derive the functional equation for $\mathcal{R}_2$.

$$\mathcal{R}_2 = \{(x, y) | 1 \leq y \leq 2x\}$$

For this second region, the slicing operator changes, so we begin again:

$$\begin{aligned}
f(z) &= z \\
\mathcal{L}[F(z,u)] &= \sum_j f^{\langle j\rangle}(z)\mathcal{L}[u^j] \\
&= \sum_j f^{\langle j\rangle}(z) u \frac{1-u^{2j}}{1-u} \\
&= \frac{u}{1-u}\sum_j f^{\langle j\rangle}(z)(1-u^{2j}) \\
&= \frac{u}{1-u}\left(\sum_j f^{\langle j\rangle}(z) - \sum_j f^{\langle j\rangle}(z)(u^2)^j\right) \\
&= \frac{u}{1-u}\big(F(z,1) - F(z,u^2)\big)
\end{aligned}$$

Hence,

$$F(z,u) = f(zu) + (\mathcal{L}[F(z,u)])_{u\mapsto zu} = zu + \frac{zu}{1-zu}\big(F(z,1) - F(z,z^2u^2)\big) \tag{3.12}$$

Repeated substitution gives:

$$F(z,u) = \alpha(u) - \frac{zu}{1-zu}\alpha(z^2u^2) + \frac{zu}{1-zu}\frac{z^3u^2}{1-z^3u^2}\alpha(z^6u^4) - \cdots$$

where

$$\alpha(u) = zu + \frac{zu}{1-zu}F(z,1), \quad \beta(u) = -\frac{zu}{1-zu}, \quad \sigma(u) = z^2u^2$$

Step 2. Solve for $F(z,1)$.

Set $u=1$ and write $X \coloneqq F(z,1)$:

$$\begin{aligned}
X &= \alpha(1) - \frac{z}{1-z}\alpha(z^2) + \frac{z}{1-z}\frac{z^3}{1-z^3}\alpha(z^6) - \cdots \\
&= \left[z + \frac{z}{1-z}X\right] - \frac{z}{1-z}\left[z^3 + \frac{z^3}{1-z^3}X\right] + \frac{z}{1-z}\frac{z^3}{1-z^3}\left[z^7 + \frac{z^7}{1-z^7}X\right] - \cdots
\end{aligned}$$

This is a linear equation in X:

$$X = \text{Series A involving z} + X \cdot \text{Series B involving z} \tag{3.13}$$

From here, separate the constant contribution from the coefficient of X. This gives

$$\begin{aligned}
\text{Series A} &= z - \frac{z}{1-z}z^3 + \frac{z}{1-z}\frac{z^3}{1-z^3}z^7 - \cdots \\
&= z - \frac{z^{1+3}}{1-z} + \frac{z^{1+3+7}}{(1-z)(1-z^3)} - \cdots \\
&= \sum_{j\geq 1}(-1)^{j-1}z^{2^{j+1}-j-2} / \Big((1-z)(1-z^3)\cdots(1-z^{2^{j-1}-1})\Big) \\
&= \sum_{j\geq 1}(-1)^{j-1}z^{2^{j+1}-j-2}/Q_{j-1}(z)
\end{aligned}$$

The exponent is:

$$\begin{aligned}
1+3+7+\cdots+(2^j-1) &= (2-1)+(2^2-1)+(2^3-1)+\cdots+(2^j-1)\\
&= 2+2^2+2^3+\cdots+2^j-j\\
&= \frac{2-2^{j+1}}{1-2}-j\\
&= 2^{j+1}-j-2
\end{aligned}$$

$$\begin{aligned}
1-\text{Series B} &= 1-\frac{z}{1-z}+\frac{z}{1-z}\frac{z^3}{1-z^3}-\frac{z}{1-z}\frac{z^3}{1-z^3}\frac{z^7}{1-z^7}+\cdots\\
&= 1-\frac{z}{1-z}+\frac{z^{1+3}}{(1-z)(1-z^3)}-\frac{z^{1+3+7}}{(1-z)(1-z^3)(1-z^7)}+\cdots\\
&= \sum_{j\geq 0}(-1)^j z^{2^{j+1}-j-2}/\Big((1-z)(1-z^3)\cdots(1-z^{2^j-1})\Big)\\
&= \sum_{j\geq 0}(-1)^j z^{2^{j+1}-j-2}/Q_j(z)
\end{aligned}$$

Finally,

$$F(z,1)=X=\frac{\sum_{j\geq 1}(-1)^{j-1}z^{2^{j+1}-j-2}/Q_{j-1}(z)}{\sum_{j\geq 0}(-1)^j z^{2^{j+1}-j-2}/Q_j(z)} \tag{3.14}$$

This conclusion belongs only to $\mathcal{R}_2$; it does not depend on the earlier computation for $\mathcal{R}_1$.

The sequence begins **1, 1, 2, 3, 5,** 9, 16, 28, 50 and is *EIS* **A002572**. It counts partitions of 1 into summands of the form $\frac{1}{2},\frac{1}{4},\frac{1}{8},\cdots$: there is **1** way to represent 1 as *2* summands; **1** way as *3* summands; **2** ways as *4* summands; **3** ways as *5* summands; **5** ways as *6* summands $\cdots$

$$1 = \frac{1}{2} + \frac{1}{2}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{4}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$
$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$
$$1 = \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$
$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$
$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$
$$1 = \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$
$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}$$
$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32}$$

$\triangleright$ **III.32.** *Carlitz compositions I.*

To obtain the functional equation, note that

$$\begin{aligned}
k(z) &= z + z^2 + \cdots = \frac{z}{1-z} \\
\mathcal{L}[K(z,u)] &= \sum_j k^{\langle j \rangle}(z)\mathcal{L}[u^j] \\
&= \sum_j k^{\langle j \rangle}(z)\left(\frac{u}{1-u} - u^j\right) \\
&= \frac{u}{1-u}\sum_j k^{\langle j \rangle}(z) - \sum_j k^{\langle j \rangle}(z)u^j \\
&= \frac{u}{1-u}K(z,1) - K(z,u)
\end{aligned}$$

Therefore,

$$K(z,u) = k(zu) + (\mathcal{L}[K(z,u)])_{u\mapsto zu} = \frac{zu}{1-zu} + \frac{zu}{1-zu}K(z,1) - K(z,zu)$$

To solve for $K(z,1)$, write $X = K(z,1)$:

$$K(z,u) = \frac{zu}{1-zu}(1+X) - K(z,zu)$$

One iteration gives:

$$K(z,u) = \frac{zu}{1-zu}(1+X) - \left(\frac{z^2u}{1-z^2u}(1+X) - K(z,z^2u)\right)$$

Iterating all the way gives:

$$\begin{aligned} K(z,u) &= \frac{zu}{1-zu}(1+X) - \frac{z^2u}{1-z^2u}(1+X) + \frac{z^3u}{1-z^3u}(1+X) - \cdots \\ &= (1+X)\sum_{j\geq 1}(-1)^{j-1}\frac{z^ju}{1-z^ju} \end{aligned}$$

Now set $u=1$:

$$X = (1+X)\sum_{j\geq 1}\frac{(-1)^{j-1}z^j}{1-z^j}$$

If we call the sum S, then

$$X = (1+X)S \implies 1+X = \frac{1}{1-S} = \left(1 - \sum_{j\geq 1}\frac{(-1)^{j-1}z^j}{1-z^j}\right)^{-1} = \left(1 + \sum_{j\geq 1}\frac{(-z)^j}{1-z^j}\right)^{-1}$$

The small bookkeeping point is that the final expression equals $1+X$, not X itself.

***Example* III.23.** *Local order patterns in permutations.*

Step 1. Solve the differential equation.

For brevity, let $\hat{I}(z,\mathbf{u})$ be abbreviated to I. Then the differential equation becomes

$$\frac{\mathrm{d}I}{\mathrm{d}z} = u_0 + (u_1 + u_1')I + u_2I^2 = u_0 + 2v_1I + u_2I^2$$

Separating variables yields:

$$\frac{\mathrm{d}I}{u_0 + 2v_1I + u_2I^2} = \mathrm{d}z$$

Integrate this from the initial condition $I(0)=0$ up to a general z:

$$\int_{\xi=I(0)}^{\xi=I(z)} \frac{\mathrm{d}\xi}{u_0 + 2v_1\xi + u_2\xi^2} = \int_{\zeta=0}^{\zeta=z} \mathrm{d}\zeta$$

Using $I(0)=0$ and writing $I(z)$ simply as I,

$$\int_0^I \frac{\mathrm{d}\xi}{u_0 + 2v_1\xi + u_2\xi^2} = \int_0^z \mathrm{d}\zeta = z$$

Complete the square in the denominator:

$$u_0 + 2v_1\xi + u_2\xi^2 = u_2\left[\left(\xi + \frac{v_1}{u_2}\right)^2 + \frac{\delta^2}{u_2^2}\right]$$

which turns the integral into

$$\int_0^I \frac{\mathrm{d}\xi}{u_2\left[\left(\xi+\frac{v_1}{u_2}\right)^2+\left(\frac{\delta}{u_2}\right)^2\right]} = z$$

Using the standard integral $\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a}$:

$$z = \frac{1}{u_2}\frac{1}{\frac{\delta}{u_2}}\arctan\left.\frac{\xi+\frac{v_1}{u_2}}{\frac{\delta}{u_2}}\right|_{\xi=0}^{\xi=I} = \frac{1}{\delta}\left[\arctan\left(\frac{u_2 I+v_1}{\delta}\right)-\arctan\left(\frac{v_1}{\delta}\right)\right]$$

Now rearrange and take tangents:

$$\frac{u_2 I+v_1}{\delta} = \tan\left[\arctan\left(\frac{v_1}{\delta}\right)+z\delta\right] = \frac{\frac{v_1}{\delta}+\tan(z\delta)}{1-\frac{v_1}{\delta}\tan(z\delta)} = \frac{v_1+\delta\tan(z\delta)}{\delta-v_1\tan(z\delta)}$$

Solving for I yields:

$$I = \frac{\delta}{u_2}\frac{v_1+\delta\tan(z\delta)}{\delta-v_1\tan(z\delta)} - \frac{v_1}{u_2} \tag{3.15}$$

Step 2. Expand the closed form.

Now expand I as a power series. Start from the closed form (3.15):

$$\begin{aligned}
I &= \frac{v_1}{u_2}\frac{1+\frac{\delta}{v_1}\tan(z\delta)}{1-\frac{v_1}{\delta}\tan(z\delta)} - \frac{v_1}{u_2}\\
&= \frac{v_1}{u_2}\left[1+\frac{\delta}{v_1}\tan(z\delta)\right]\left[1+\frac{v_1}{\delta}\tan(z\delta)+\left(\frac{v_1}{\delta}\tan(z\delta)\right)^2+\left(\frac{v_1}{\delta}\tan(z\delta)\right)^3+\cdots\right]-\frac{v_1}{u_2}\\
&= \frac{v_1}{u_2}\left[1+\frac{\delta}{v_1}\left(z\delta+\frac{(z\delta)^3}{3}+\cdots\right)\right]\left[1+\frac{v_1}{\delta}\left(z\delta+\frac{(z\delta)^3}{3}\right)+\left(\frac{v_1}{\delta}(z\delta)\right)^2+\left(\frac{v_1}{\delta}(z\delta)\right)^3+\cdots\right]-\frac{v_1}{u_2}\\
&= \frac{v_1}{u_2}\left[1+\frac{\delta^2}{v_1}z+\frac{\delta^4}{3v_1}z^3+\cdots\right]\left[1+v_1 z+v_1^2 z^2+\left(\frac{v_1\delta^2}{3}+v_1^3\right)z^3+\cdots\right]-\frac{v_1}{u_2}\\
&= u_0 z+u_0(u_1+u_1')\frac{z^2}{2!}+u_0((u_1+u_1')^2+2u_0u_2)\frac{z^3}{3!}+\cdots
\end{aligned}$$

The same coefficients could also be recovered directly from the differential equation by repeated differentiation, but the closed form is shorter.

Step 3. Check the first few objects directly.

There is also a direct check: list the objects for small n.

$$\begin{aligned}\mathcal{P}_1 &= \{①\} \implies \\ f_1(\mathbf{u}) &= u_0\end{aligned}$$

$$\begin{aligned}\mathcal{P}_2 &= \{①\text{-}②, ②\text{-}①\} \implies \\ f_2(\mathbf{u}) &= u_1u_0 + u_0u_1' = u_0(u_1 + u_1')\end{aligned}$$

$$\begin{aligned}\mathcal{P}_3 &= \{①\text{-}②\text{-}③, ①\text{-}③\text{-}②, ②\text{-}①\text{-}③, ②\text{-}③\text{-}①, ③\text{-}①\text{-}②, ③\text{-}②\text{-}①\} \implies \\ f_3(\mathbf{u}) &= u_1u_1u_0 + u_1u_0u_1' + u_0u_2u_0 + u_1u_0u_1' + u_0u_2u_0 + u_0u_1'u_1' \\ &= u_0(u_1^2 + u_1u_1' + u_0u_2 + u_1u_1' + u_0u_2 + u_1'^2) \\ &= u_0((u_1 + u_1')^2 + 2u_0u_2)\end{aligned}$$

$$\begin{aligned}\mathcal{P}_4 &= \{\cdots\} \\ f_4(\mathbf{u}) &= u_0(u_1 + u_1')((u_1 + u_1')^2 + 8u_0u_2)\end{aligned}$$

With the closed form established, we can now extract the mean numbers of the three node types by specializing one marker at a time.

Nullary Type

Set $u_1 = u_1' = u_2 = 1$. Then $v_1 = 1$ and $\delta = \sqrt{u_0 - 1}$. We have

$$\hat{I}(z, u_0) = \delta\frac{1 + \delta\tan(z\delta)}{\delta - \tan(z\delta)} - 1 = (1 + \delta^2)\frac{w}{1 - w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. Write $w = w_0 + w_2\delta^2 + O(\delta^4)$ with $w_0 = z$ and $w_2 = z^3/3$. Then

$$1 - w = 1 - w_0 - w_2\delta^2 + O(\delta^4) = (1 - w_0)\left[1 - \frac{w_2}{1 - w_0}\delta^2 + O(\delta^4)\right]$$

Hence,

$$\frac{1}{1 - w} = \frac{1}{1 - w_0}\frac{1}{1 - \frac{w_2}{1 - w_0}\delta^2 + \cdots} = \frac{1}{1 - w_0}\left[1 + \frac{w_2}{1 - w_0}\delta^2 + O(\delta^4)\right]$$

Multiplying by $w = w_0 + w_2\delta^2 + O(\delta^4)$ yields:

$$\frac{w}{1 - w} = \frac{w_0}{1 - w_0} + \frac{w_2}{(1 - w_0)^2}\delta^2 + O(\delta^4)$$

Substituting $w_0 = z$ and $w_2 = z^3/3$, we obtain

$$\frac{w}{1 - w} = \frac{z}{1 - z} + \frac{z^3}{3(1 - z)^2}\delta^2 + O(\delta^4) \tag{3.16}$$

Since $\delta^2 = u_0 - 1$, the first-order Taylor term in u_0 is exactly the coefficient of δ^2. Hence,

$$\left.\frac{\partial I}{\partial u_0}\right|_{u_0=1} = [\delta^2]\left((1 + \delta^2)\frac{w}{1 - w}\right) = \frac{z}{1 - z} + \frac{z^3}{3(1 - z)^2}$$

The coefficient of z^n in $\frac{\partial I}{\partial u_0}\Big|_{u_0=1}$ is $(n+1)/3$.

Unary Type

Treat the two unary types together. Note that $\hat{I}$ depends on u_1, u_1' only through $v_1 = (u_1 + u_1')/2$. Set $u_0 = u_2 = 1$. Then $\delta = \sqrt{1 - v_1^2}$ and

$$\hat{I}(z, v_1) = \delta \frac{v_1 + \delta \tan(z\delta)}{\delta - v_1 \tan(z\delta)} - v_1 = \frac{v_1 + \delta^2 w}{1 - v_1 w} - v_1 = \frac{w}{1 - v_1 w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. We have

$$\frac{\partial I}{\partial u_1} = \frac{\partial I}{\partial u_1'} = \frac{1}{2}\frac{\partial I}{\partial v_1}$$

So it is enough to compute $\frac{\partial I}{\partial v_1}$ at $v_1 = 1$. First, w depends on v_1 through $s := \delta^2 = 1 - v_1^2$:

$$\frac{\partial w}{\partial v_1} = \frac{\partial w}{\partial s} \cdot (-2v_1)$$

From the series $\frac{\partial w}{\partial s}\Big|_{s=0} = \frac{z^3}{3}$, so at $v_1 = 1$:

$$\frac{\partial w}{\partial v_1}\Big|_{v_1=1} = -\frac{2z^3}{3}$$

Now differentiate $\hat{I}(z, v_1)$:

$$\frac{\partial \hat{I}}{\partial v_1} = \frac{\frac{\partial w}{\partial v_1}(1 - v_1 w) + w(w + v_1 \frac{\partial w}{\partial v_1})}{(1 - v_1 w)^2}$$

Substituting $v_1 = 1, w = z$ and $\frac{\partial w}{\partial v_1} = -\frac{2z^3}{3}$, we obtain

$$\frac{\partial \hat{I}}{\partial v_1}\Big|_{v_1=1} = \frac{z^2\left(1 - \frac{2}{3}z\right)}{(1-z)^2}$$

Consequently

$$\frac{\partial \hat{I}}{\partial u_1}\Big|_{u_1=1} = \frac{\partial \hat{I}}{\partial u_1'}\Big|_{u_1'=1} = \frac{z^2\left(1 - \frac{2}{3}z\right)}{2(1-z)^2}$$

The coefficient of z^n in $\frac{\partial \hat{I}}{\partial u_1}\Big|_{u_1=1}$ and $\frac{\partial \hat{I}}{\partial u_1'}\Big|_{u_1'=1}$ is $(n+1)/6$.

Binary Type

Set $u_0 = 1$, $v_1 = (1+1)/2 = 1$. Then $\delta = \sqrt{u_2 - 1}$ and

$$\hat{I}(z, u_2) = \frac{\delta}{u_2}\frac{1 + \delta \tan(z\delta)}{\delta - \tan(z\delta)} - \frac{1}{u_2} = \frac{w}{1 - w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. This is the same calculation as in the nullary case, with one parameter renamed. From 3.16 we obtain

$$\frac{\partial \hat{I}}{\partial u_2}\Big|_{u_2=1} = [\delta^2]\frac{w}{1-w} = \frac{z^3}{3(1-z)^2}$$

The coefficient of z^n in $\left.\frac{\partial \hat{I}}{\partial u_2}\right|_{u_2=1}$ is $\frac{n-2}{3}$.

Summing the contributions of all node types:

$$\frac{n+1}{3} + \frac{n+1}{6} + \frac{n+1}{6} + \frac{n-2}{3} = n$$

as expected.

III.7.4. *Symbolic inclusion-exclusion.*

To make the relation $\mathcal{Q} \cong \mathcal{U} \star \mathcal{P}$ concrete, let us list the elements of $\mathcal{Q}_3$.

$\mathcal{U} = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}, \cdots\}$
$\mathcal{P} = \{\emptyset, (1), (1,2), (2,1), (1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1), \cdots\}$

Pick an element of $\mathcal{U}$ of size k and an element of $\mathcal{P}$ of size $3-k$, with $0 \leq k \leq 3$.

$\mathcal{U}_0 \star \mathcal{P}_3$

- $\emptyset \star (1,2,3)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (1,2,3)) \leftrightarrow (1,2,3)$
- $\emptyset \star (1,3,2)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (1,3,2)) \leftrightarrow (1,3,2)$
- $\emptyset \star (2,1,3)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (2,1,3)) \leftrightarrow (2,1,3)$
- $\emptyset \star (2,3,1)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (2,3,1)) \leftrightarrow (2,3,1)$
- $\emptyset \star (3,1,2)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (3,1,2)) \leftrightarrow (3,1,2)$
- $\emptyset \star (3,2,1)$: $\binom{3}{0} = 1$ way of relabelling
  - $(\emptyset, (3,2,1)) \leftrightarrow (3,2,1)$

$\mathcal{U}_1 \star \mathcal{P}_2$

- $\{1\} \star (1,2)$: $\binom{3}{1} = 3$ ways of relabelling
  - $(\{1\}, (2,3)) \leftrightarrow (\underline{1}, 2, 3)$
  - $(\{2\}, (1,3)) \leftrightarrow (1, \underline{2}, 3)$
  - $(\{3\}, (1,2)) \leftrightarrow (1, 2, \underline{3})$
- $\{1\} \star (2,1)$: $\binom{3}{1} = 3$ ways of relabelling

  - $(\{1\},(3,2)) \leftrightarrow (\underline{1},3,2)$
  - $(\{2\},(3,1)) \leftrightarrow (3,\underline{2},1)$
  - $(\{3\},(2,1)) \leftrightarrow (2,1,\underline{3})$

$\mathcal{U}_2 \star \mathcal{P}_1$

- $\{1,2\} \star (1)$: $\binom{3}{2} = 3$ ways of relabelling
  - $(\{1,2\},(3)) \leftrightarrow (\underline{1},\underline{2},3)$
  - $(\{1,3\},(2)) \leftrightarrow (\underline{1},2,\underline{3})$
  - $(\{2,3\},(1)) \leftrightarrow (1,\underline{2},\underline{3})$

$\mathcal{U}_3 \star \mathcal{P}_0$

- $\{1,2,3\} \star \emptyset$: $\binom{3}{3} = 1$ way of relabelling
  - $(\{1,2,3\},\emptyset) \leftrightarrow (\underline{1},\underline{2},\underline{3})$

***Example* III.25.** *Rises and ascending runs in permutations.*

Rises

Step 1. Expand around $z = 0$.

First expand $A(z,u)$ as a power series in z around 0.

$$A(z,u) = \frac{u-1}{u-e^{z(u-1)}}$$

The denominator is:

$$\begin{aligned}
u - e^{z(u-1)} &= u - \left[1 + (u-1)z + \frac{(u-1)^2}{2}z^2 + \frac{(u-1)^3}{6}z^3 + \frac{(u-1)^4}{24}z^4 + O(z^5)\right] \\
&= (u-1) - (u-1)z - \frac{(u-1)^2}{2}z^2 - \frac{(u-1)^3}{6}z^3 - \frac{(u-1)^4}{24}z^4 + O(z^5) \\
&= (u-1)\left[1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4 + O(z^5)\right]
\end{aligned}$$

Hence,

$$\begin{aligned}
A(z,u) &= \left(1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4 + O(z^5)\right)^{-1} \\
&= \left(1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4\right)^{-1} + O(z^5)
\end{aligned}$$

Set

$$w = z + \frac{u-1}{2}z^2 + \frac{(u-1)^2}{6}z^3 + \frac{(u-1)^3}{24}z^4$$

Compute powers of w up to order z^4:

$$\begin{aligned} w^2 &= \left(z + \frac{u-1}{2}z^2 + \frac{(u-1)^2}{6}z^3\right)^2 \\ &= z^2 + \left(\frac{u-1}{2}z^2\right)^2 + 2z\frac{u-1}{2}z^2 + 2z\frac{(u-1)^2}{6}z^3 \\ &= z^2 + (u-1)z^3 + \frac{7}{12}(u-1)^2 z^4 \end{aligned}$$

$$w^3 = \left(z + \frac{u-1}{2}z^2\right)^3 = z^3 + 3z^2\frac{u-1}{2}z^2 = z^3 + \frac{3}{2}(u-1)z^4$$

$$w^4 = z^4$$

Then

$$\begin{aligned} A(z,u) = 1 + z &+ \left(\frac{u-1}{2} + 1\right) z^2 + \left(\frac{(u-1)^2}{6} + (u-1) + 1\right) z^3 \\ &+ \left(\frac{(u-1)^3}{24} + \frac{7}{12}(u-1)^2 + \frac{3}{2}(u-1) + 1\right) z^4 + O(z^5) \\ &= 1 + z + (u+1)\frac{z^2}{2} + (u^2+4u+1)\frac{z^3}{6} + (u^3+11u^2+11u+1)\frac{z^4}{24} + \cdots \end{aligned}$$

Step 2. Expand around $u = 1$.

Now expand $A(z,u)$ in powers of u around $u = 1$. Set $v = u - 1$.

$$A(z,u) = \frac{v}{(v+1) - e^{zv}}$$

The denominator

$$v + 1 - e^{zv} = v + 1 - \left(1 + zv + \frac{z^2}{2}v^2 + \frac{z^3}{6}v^3 + O(v^4)\right) = (1-z)v - \frac{z^2}{2}v^2 - \frac{z^3}{6}v^3 + O(v^4)$$

Hence,

$$\begin{aligned} A(z,u) &= \left(1 - z - \frac{z^2}{2}v - \frac{z^3}{6}v^2 + O(v^3)\right)^{-1} \\ &= \frac{1}{1-z}\left(1 - \frac{z^2}{2(1-z)}v - \frac{z^3}{6(1-z)}v^2 + O(v^3)\right)^{-1} \\ &= \frac{1}{1-z}\left(1 - \frac{z^2}{2(1-z)}v - \frac{z^3}{6(1-z)}v^2\right)^{-1} + O(v^3) \\ &= \frac{1}{1-z}\left(1 + \frac{z^2}{2(1-z)}v + \frac{z^3}{6(1-z)}v^2 + \left(\frac{z^2}{2(1-z)}v\right)^2\right) + O(v^3) \\ &= \frac{1}{1-z} + \frac{z^2}{2(1-z)^2}v + \frac{2z^3+z^4}{6(1-z)^3}\frac{v^2}{2!} + \cdots \end{aligned}$$

Step 3. Extract the mean and variance.

Extracting coefficients gives:

$$[z^n]\frac{z^2}{2(1-z)^2} = \frac{1}{2}[z^{n-2}]\sum_{m=0}^{\infty}(m+1)z^m = \frac{n-1}{2}$$

$$[z^n]\frac{2z^3+z^4}{6(1-z)^3} = \frac{1}{3}[z^{n-3}]\sum_{m=0}^{\infty}\binom{m+2}{2}z^m + \frac{1}{6}[z^{n-4}]\sum_{m=0}^{\infty}\binom{m+2}{2}z^m = \frac{3n^2-11n+10}{12}$$

The mean of the number of rises in a random permutation of size n is $\frac{n-1}{2}$, and the variance is:

$$\frac{3n^2-11n+10}{12} + \frac{n-1}{2} - \left(\frac{n-1}{2}\right)^2 = \frac{n+1}{12}$$

Ascending Runs

Step 1. Decompose the rational form.

The ordinary generating function $I(z,v)$ is an improper rational function: the numerator has degree $l+1$ and the denominator has degree l. By polynomial long division and standard partial fraction decomposition, $I(z,v)$ can be written exactly as:

$$I(z,v) = 1 - z + \sum_{j=1}^{l}\frac{c_j}{1-\omega_j z}$$

The polynomial part is $1-z$ (or more precisely, $-z-\frac{1}{v}$ when $l=1$). This suggests rechecking the factor $(1-z)(v+1)$ in the text, because the computation here points instead to the polynomial part stated above.

Step 2. Estimate the moments.

The mean number of ascending runs of length $l-1$ is $\frac{n-l+1}{l!} \sim \frac{n}{l!}$; the variance is $\sim \frac{n}{l!}$. Concentration holds provided

$$\lim_{n\to\infty}\frac{\sigma_n}{\mu_n} = 0$$

Now

$$\left(\frac{\sigma_n}{\mu_n}\right)^2 = \frac{\sigma_n^2}{\mu_n^2} = \frac{\frac{n}{l!}}{\left(\frac{n}{l!}\right)^2} = \frac{l!}{n}$$

This tends to 0 whenever $l! = o(n)$. So the condition $l! = o(n)$ already suffices for concentration; if the text writes $(l+1)! = o(n)$, that stronger condition is not needed for the estimate derived here.

▷ **III.36.** *Permutations without l-ascending runs.*

Consider the denominator of $A(z,u)|_{u=0}$ before applying the transform:

$$1 - z - I(z,-1) = \frac{1}{1+z+\cdots+z^l} = \frac{1-z}{1-z^{l+1}} = \frac{1-z}{1-z^m}$$

where $m := l+1$. To prepare for the transform $(1-\omega z)^{-1} \mapsto e^{\omega z}$, we decompose $\frac{1-z}{1-z^m}$ using the m-th root of unity. Let $\omega_k = e^{2\pi i k/m}$ for $k = 0, 1, \cdots, m-1$ be the roots of

$z^m = 1$. The standard partial fraction expansion is:

$$\frac{1}{1-z^m} = \frac{1}{m}\sum_{k=0}^{m-1}\frac{1}{1-\omega_k z}$$

Now apply the transform:

$$\begin{aligned}
\text{EGF Denominator} &= (1-z)\widehat{\frac{1}{1-z^m}} \\
&= \widehat{\frac{1}{1-z^m}} - \widehat{\frac{z}{1-z^m}} \\
&= \widehat{\frac{1}{1-z^m}} - \int_0^z \widehat{\frac{1}{1-t^m}}\,\mathrm{d}t \\
&= \frac{1}{m}\sum_{k=0}^{m-1} e^{\omega_k z} - \int_0^z \frac{1}{m}\sum_{k=0}^{m-1} e^{\omega_k t}\,\mathrm{d}t \\
&= \frac{1}{m}\sum_{k=0}^{m-1}\sum_{n\geq 0}\omega_k^n\frac{z^n}{n!} - \int_0^z \frac{1}{m}\sum_{k=0}^{m-1}\sum_{n\geq 0}\omega_k^n\frac{t^n}{n!}\,\mathrm{d}t \\
&= \frac{1}{m}\sum_{n\geq 0}\left(\sum_{k=0}^{m-1}\omega_k^n\right)\frac{z^n}{n!} - \int_0^z \frac{1}{m}\sum_{n\geq 0}\left(\sum_{k=0}^{m-1}\omega_k^n\right)\frac{t^n}{n!}\,\mathrm{d}t \\
&= \sum_{i\geq 0}\frac{z^{m\cdot i}}{(m\cdot i)!} - \int_0^z \sum_{i\geq 0}\frac{t^{m\cdot i}}{(m\cdot i)!}\,\mathrm{d}t \\
&= \sum_{i\geq 0}\frac{z^{m\cdot i}}{(m\cdot i)!} - \frac{z^{m\cdot i+1}}{(m\cdot i+1)!}
\end{aligned}$$

At this point the root-of-unity filter has done its job: the denominator is now a sparse exponential series that can be read term by term. The key points are these: multiplying by z in the ordinary generating-function setting becomes an integral after passage to exponential generating functions, and $\sum_{k=0}^{m-1}\omega_k^n$ vanishes unless n is a multiple of m, in which case it equals m.

For orientation, here are the first few EGFs when $m = 2, 3, 4$:

$$\text{A.R. of length 1} \quad \left(\sum_{i\geq 0}\frac{z^{2i}}{(2i)!} - \frac{z^{2i+1}}{(2i+1)!}\right)^{-1} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + O(z^5)$$

$$\text{A.R. of length 2} \quad \left(\sum_{i\geq 0}\frac{z^{3i}}{(3i)!} - \frac{z^{3i+1}}{(3i+1)!}\right)^{-1} = 1 + z + 2\frac{z^2}{2!} + 5\frac{z^3}{3!} + 17\frac{z^4}{4!} + O(z^5)$$

$$\text{A.R. of length 3} \quad \left(\sum_{i\geq 0}\frac{z^{4i}}{(4i)!} - \frac{z^{4i+1}}{(4i+1)!}\right)^{-1} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + 23\frac{z^4}{4!} + O(z^5)$$

For example, there are 17 permutations in $\mathcal{P}_4$ without ascending runs of length 2: 1324, 1423, 1432, 2143, 2314, 2413, 2431, 3142, 3214, 3241, 3412, 3421, 4132, 4213, 4231, 4312, 4321.

$\triangleright$ **III.41.** *Patterns in trees II.*

We want to show

$$u^{[\![\tau=\mathfrak{t}]\!]}u^{\omega[\tau_0]}u^{\omega[\tau_1]} = u^{\omega[\tau_0]}u^{\omega[\tau_1]} + [\![\tau = \mathfrak{t}]\!] \cdot (u-1)$$

- $\tau \neq \mathfrak{t}$: LHS $= u^0 u^{\omega[\tau_0]}u^{\omega[\tau_1]} = u^{\omega[\tau_0]}u^{\omega[\tau_1]}$; RHS $= u^{\omega[\tau_0]}u^{\omega[\tau_1]} + 0 \cdot (u-1) = u^{\omega[\tau_0]}u^{\omega[\tau_1]}$
- $\tau = \mathfrak{t}$: LHS $= u^1 u^0 u^0 = u$; RHS $= u^0 u^0 + 1 \cdot (u-1) = u$

Now sum over all binary trees τ to get the bivariate generating function:

$$\begin{aligned}
B(z,u) &= \sum_{\tau} z^{|\tau|}u^{\omega[\tau]} \\
&= z^{|\emptyset|}u^{\omega[\emptyset]} + \sum_{\tau\neq\emptyset} z^{|\tau|}u^{\omega[\tau]} \\
&= z^0u^0 + \sum_{\tau\neq\emptyset} z^{|\tau|}u^{\omega[\tau]} \\
&= 1 + \sum_{\tau\neq\emptyset} z^{|\tau|}\left(u^{\omega[\tau_0]}u^{\omega[\tau_1]} + [\![\tau = \mathfrak{t}]\!] \cdot (u-1)\right) \\
&= 1 + \sum_{\tau_0,\tau_1} z^{1+|\tau_0|+|\tau_1|}u^{\omega[\tau_0]}u^{\omega[\tau_1]} + \sum_{\tau\neq\emptyset} z^{|\tau|}[\![\tau = \mathfrak{t}]\!] \cdot (u-1) \\
&= 1 + z\sum_{\tau_0,\tau_1} z^{|\tau_0|}u^{\omega[\tau_0]} \cdot z^{|\tau_1|}u^{\omega[\tau_1]} + z^{|\mathfrak{t}|} \cdot (u-1) \\
&= 1 + zB(z,u)^2 + z^m(u-1)
\end{aligned}$$

3.8 Extremal parameters

III.8.3. ***Averages and moments.***

To see why the first sum rearranges into the second, use a discrete summation-by-parts trick. Write the linear factor h as

$$h = \sum_{j=1}^{h} 1$$

Substituting this produces a double sum:

$$\Xi(z) = \sum_{h=0}^{\infty}\left(\sum_{j=1}^{h} 1\right)\left[f^{[h]}(z) - f^{[h-1]}(z)\right] = \sum_{h=1}^{\infty}\sum_{j=1}^{h}\left[f^{[h]}(z) - f^{[h-1]}(z)\right]$$

The summation region is $1 \leq j \leq h < \infty$. Reverse the order: then j runs from 1 to ∞, while h runs from j to ∞.

$$\Xi(z) = \sum_{j=1}^{\infty}\sum_{h=j}^{\infty}\left[f^{[h]}(z) - f^{[h-1]}(z)\right]$$

The inner sum over h now telescopes:

$$\sum_{h=j}^{\infty}\left[f^{[h]}(z) - f^{[h-1]}(z)\right] = \lim_{N\to\infty}\sum_{h=j}^{N}\left[f^{[h]}(z) - f^{[h-1]}(z)\right]$$

Expanding the finite sum makes the cancellation explicit:

$$\lim_{N\to\infty}\left(\left[f^{[j]} - f^{[j-1]}\right] + \left[f^{[j+1]} - f^{[j]}\right] + \cdots + \left[f^{[N]} - f^{[N-1]}\right]\right) = \lim_{N\to\infty}\left(f^{[N]}(z) - f^{[j-1]}(z)\right)$$

Because $f^{[N]}(z)$ counts objects with parameter at most N, letting $N \to \infty$ recovers the unconstrained generating function $f(z)$. So the inner sum collapses to

$$f(z) - f^{[j-1]}(z)$$

Putting this back into the outer sum,

$$\Xi(z) = \sum_{j=1}^{\infty}\left[f(z) - f^{[j-1]}(z)\right]$$

Finally shift the index by $h = j - 1$:

$$\Xi(z) = \sum_{h=0}^{\infty}\left[f(z) - f^{[h]}(z)\right]$$

3.9 Perspective

Chapter 4

COMPLEX ANALYSIS, RATIONAL AND MEROMORPHIC ASYMPTOTICS

4.1 Generating functions as analytic objects

▷ **IV.6.** *Integrals of rational fractions.*

$$J_m = \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{(1+x^2)^m}$$

Set

$$f(z) = \frac{1}{(1+z^2)^m} = \frac{1}{(z-i)^m(z+i)^m}.$$

Integrate over the contour Γ_R formed by the real interval $[-R, R]$ and the upper semicircle $C_R = \{z = Re^{i\theta} : 0 \le \theta \le \pi\}$.

For $z \in C_R$ and $R > 1$, $|z^2+1| \ge |z^2| - 1 = R^2 - 1$. Thus

$$|f(z)| \le \frac{1}{(R^2-1)^m}$$

The length of C_R is πR, so

$$\left| \int_{C_R} f(z)\,\mathrm{d}z \right| \le \frac{\pi R}{(R^2-1)^m} \to 0 \text{ as } R \to \infty$$

So the contour integral reduces to a residue computation at the pole in the upper half-plane. Therefore

$$\lim_{R\to\infty} \int_{\Gamma_R} f(z)\,\mathrm{d}z = \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{(1+x^2)^m} = J_m$$

For $R > 1$, the only singularity inside Γ_R is the pole of order m at $z = i$. The residue theorem then gives:

$$\int_{\Gamma_R} f(z)\,\mathrm{d}z = 2\pi i \operatorname{Res}(f; i)$$

Now compute the residue at $z = i$:

$$\mathrm{Res}(f;i) = \frac{1}{(m-1)!}\lim_{z\to i}\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}[(z-i)^m f(z)] = \frac{1}{(m-1)!}\lim_{z\to i}\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}(z+i)^{-m}$$

The kth derivative of $g(z) \coloneqq (z+i)^{-m}$ is:

$$g^{(k)}(z) = (-1)^k\frac{(m+k-1)!}{(m-1)!}(z+i)^{-m-k}$$

For $k = m-1$:

$$g^{(m-1)}(z) = (-1)^{m-1}\frac{(2m-2)!}{(m-1)!}(z+i)^{-2m+1}$$

Evaluating at $z = i$ gives

$$g^{(m-1)}(i) = (-1)^{m-1}\frac{(2m-2)!}{(m-1)!}(2i)^{-2m+1}$$

Therefore,

$$\begin{aligned}\mathrm{Res}(f;i) &= \frac{1}{(m-1)!}(-1)^{m-1}\frac{(2m-2)!}{(m-1)!}(2i)^{-2m+1}\\ &= (-1)^{m-1}\frac{(2m-2)!}{[(m-1)!]^2}\frac{(-1)^m i}{2^{2m-1}}\\ &= -i\frac{(2m-2)!}{[(m-1)!]^2 2^{2m-1}}\end{aligned}$$

$$J_m = 2\pi i\left(-i\frac{(2m-2)!}{[(m-1)!]^2 2^{2m-1}}\right) = \frac{\pi}{2^{2m-2}}\binom{2m-2}{m-1} = \frac{\pi}{4^{m-1}}\binom{2m-2}{m-1} \tag{4.1}$$

$$K_m = \int_{-\infty}^{\infty}\frac{\mathrm{d}x}{(1^2+x^2)(2^2+x^2)\cdots(m^2+x^2)}$$

Now set

$$f(z) = \frac{1}{\prod_{k=1}^m (z^2+k^2)}.$$

We use the same contour Γ_R.

For $R > m$, the poles inside Γ_R are $z_k = ik, k = 1, \cdots m$. On C_R, $|z^2+k^2| \geq R^2-k^2 \geq (R/2)^2$ eventually, so $|f(z)| \leq \mathrm{const}/R^{2m}$ while the length of C_R is πR. Hence,

$$\lim_{R\to\infty}\int_{C_R} f(z)\,\mathrm{d}z = 0$$

and the residue theorem yields

$$K_m = 2\pi i\sum_{k=1}^m \mathrm{Res}(f;ik)$$

Since the poles are simple,

$$\mathrm{Res}(f;ik) = \lim_{z\to ik}(z-ik)f(z) = \frac{1}{2ik}\prod_{j\neq k}\frac{1}{(ik)^2+j^2} = \frac{1}{2ik}\prod_{j\neq k}\frac{1}{j^2-k^2}$$

Therefore,

$$K_m = 2\pi i \sum_{k=1}^{m} \frac{1}{2ik} \prod_{j\neq k} \frac{1}{j^2-k^2} = \pi \sum_{k=1}^{m} \frac{1}{k} \prod_{j\neq k} \frac{1}{j^2-k^2}$$

The analytic work is now done; what remains is straightforward algebra, though it takes a little patience. Factor the product:

$$\prod_{j\neq k} j^2 - k^2 = \prod_{j\neq k}(j-k) \prod_{j\neq k}(j+k)$$

For fixed k,

$$\prod_{j\neq k}(j-k) = (1-k)(2-k)\cdots((k-1)-k)(1)(2)\cdots(m-k) = (-1)^{k-1}(k-1)!(m-k)!$$

$$\prod_{j\neq k}(j+k) = (1+k)(2+k)\cdots((k-1)+k)(2k+1)(2k+2)\cdots(m+k) = \frac{(m+k)!}{k!(2k)}$$

Therefore,

$$\frac{1}{k}\prod_{j\neq k}\frac{1}{j^2-k^2} = \frac{1}{k}\left((-1)^{k-1}(k-1)!(m-k)! \cdot \frac{(m+k)!}{k!(2k)}\right)^{-1} = \frac{(-1)^{k-1}2k}{(m-k)!(m+k)!}$$

and

$$K_m = \pi \sum_{k=1}^{m} \frac{(-1)^{k-1}2k}{(m-k)!(m+k)!} = \frac{2\pi}{(2m)!}\sum_{k=1}^{m}(-1)^{k-1}k\binom{2m}{m-k}$$

To simplify the sum

$$S := \sum_{k=1}^{m}(-1)^{k-1}k\binom{2m}{m+k},$$

generating functions are again the quickest tool. The aim is to turn the alternating sum into a single coefficient extraction. Recall that the coefficient of z^m in a product $A(z)B(z)$ is

$$[z^m]A(z)B(z) = \sum_k a_k b_{m-k}$$

Take

$$A(z) = \sum_{k=1}^{m}(-1)^{k-1}kz^k, \qquad B(z) = (1+z)^{2m} = \sum_{j=0}^{2m}\binom{2m}{j}z^j.$$

Then the coefficient of z^m in $A(z)B(z)$ is exactly S:

$$S = [z^m]\sum_{k=1}^{m}(-1)^{k-1}kz^k(1+z)^{2m} \tag{4.2}$$

Since we only care about the coefficient of z^m, terms of degree $> m$ in $A(z)$ are irrelevant. So we may extend the sum to infinity without changing the coefficient:

$$A(z) \equiv \sum_{k=1}^{\infty}(-1)^{k-1}kz^k = \frac{z}{(1+z)^2} \mod z^{m+1}$$

Hence, by (4.2),

$$S = [z^m]\frac{z}{(1+z)^2} \cdot (1+z)^{2m} = [z^m]z(1+z)^{2m-2} = \binom{2m-2}{m-1}$$

Finally,

$$K_m = \frac{2\pi}{(2m)!}\binom{2m-2}{m-1} = \frac{\pi}{m(2m-1)[(m-1)!]^2} \tag{4.3}$$

4.2 Analytic functions and meromorphic functions

4.3 Singularities and exponential growth of coefficients

4.4 Closure properties and computable bounds

4.5 Rational and meromorphic functions

4.6 Localization of singularities

4.7 Singularities and functional equations

4.8 Perspective